

STRAY FEATHERS
FROM A BIRD MAN'S DESK

AUSTIN L. RAND

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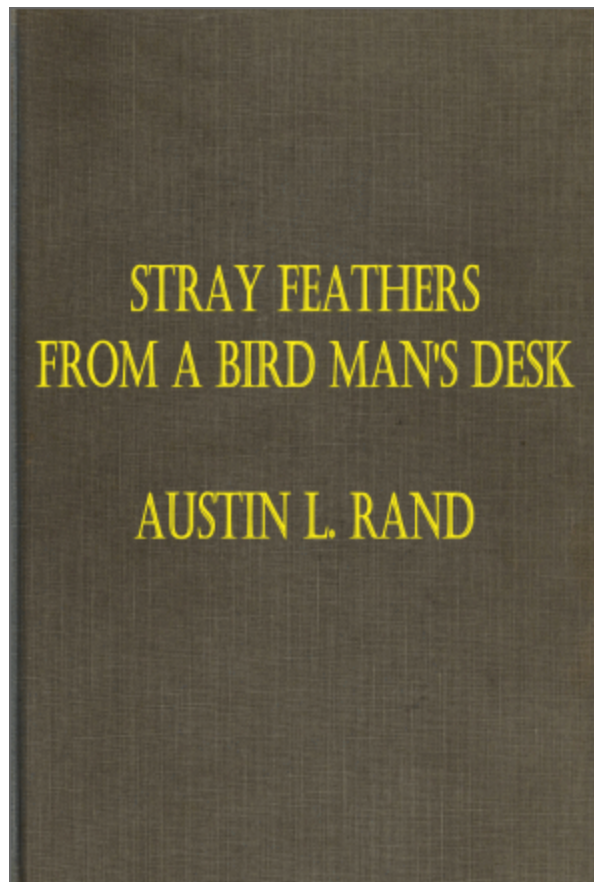
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STRAY FEATHERS FROM A BIRD MAN'S DESK

**STRAY
FEATHERS
FROM
A BIRD MAN'S
DESK**

By Austin L. Rand
CURATOR OF BIRDS,
CHICAGO NATURAL HISTORY MUSEUM

WITH CARTOONS BY RUTH JOHNSON

DOUBLEDAY & COMPANY, INC., GARDEN CITY, N.Y., 1955

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INTRODUCTION

In looking back over the preparation of these sketches I feel as though each evening I'd gathered up the bits and pieces left over from the day's work and fashioned them into designs for my own amusement and the edification of my family. Truly it's as though I'd used stray feathers, fallen from the bird skins I'd handled, and fitted them together into something of wider interest than the original.

Much of my work now is museum research, working with bird specimens and books. In fashioning a research paper I always amass a great deal more material, that is to say, information and ideas, than I am able to use in it. In place of a lumber room I have a set of files with index headings that range from Abundance and Age, through such headings as Beauty, Feathering of Feet, Fictitious, Hysteria, Pterylography, Social, Song, Tail Feathers, Valentine's Day, to Zoogeography. Here I put the information that is irrelevant at the moment but too interesting to discard. Its source is varied. Some has been accumulated while studying specimens from localities as geographically separated as Alaska, El Salvador, Gabon, Tristan da Cunha, Nepal, Negros, and New Guinea; and while writing papers that range from describing new species to discussing secondary sexual characters and ecological competition. Some have been recorded while in the field on expeditions, trips that ranged from two years in Madagascar, three expeditions in New Guinea, and a season in the Philippines to trips nearer home from the Yukon to Nova Scotia, Florida, and Central America.

Gradually information builds up under each heading, and new ones are added. These items are too interesting to remain buried in the files. They are things people want to know about. So I began to draft them into articles for publication in the museum's monthly, *The Chicago Natural History Museum Bulletin*. The response was gratifying. The press picked them up and reprinted them. One was used in a *Chicago Tribune* editorial. Several were used in commercial radio programs. Encouraged, I prepared more, soon overrunning the space available in the bulletin.

Most scientific papers are not written to be read for enjoyment. Conciseness as well as clarity are striven for, conveying certain information in a small compass. The correlations made are often obscure ones, appreciated only by scientists. Yet the material they contain is often intensely interesting, and if these papers were written in a more leisurely style, with more general correlations pointed out, they would provide both interesting and entertaining reading. In a few cases my own research falls in this class, and I've rewritten some of my own papers with this in mind (see "Battle of the Sexes and Its Evolutionary Significance").

This collection of articles, if it were a painting, could be called a conversation piece. Or it might be compared to a well-filled whatnot. Each of the sixty chapters is an independent unit, illustrating some facet of birds, their behavior, or our study of them. Some of the facts may seem unusual or bizarre, but most of them are well known and well documented. The thing that is new, if there is anything new, is the setting in which I've placed them, the manner in which I've looked at them. Taken as a whole, they touch on many different birds from many different places in their less widely known aspects, and with a human interest slant.

"But what will your professional colleagues say?" asked a friend as he flipped through the cartoons. "These pictures don't approach the subject in a very serious manner." Quite true. But a discipline must be very lightly rooted indeed if it can't stand a few caricatures and cartoons and perhaps be the better for them.

The knowledge of most people about the hornbills of tropical Africa, the gulls of Australia, the penguins of Antarctica, and the crocodile birds of the Nile is probably pretty vague. To give a frame of reference in a biological sense is impractical in the compass of one slim volume.

But a ready-made frame of reference already exists: the parallels in bird and in human. These I have used. But in so doing I am not imputing human motives and attributes to birds. The actions are similar. The workings of the human mind I understand only vaguely; that of the bird I can study only through the actions of the birds. One set of behavior may be learned and rational, one rigidly innate, entirely instinctive, and inherited, or at most modified by experience. Be that as it may, the similarity in the end result in two such different vertebrate animals as man and bird when faced with

similar problems is often close. Perhaps it is because the solutions are necessarily few; perhaps, and I incline to this feeling, it helps illustrate one aspect of the close relationship between all animate nature.

This series of articles is intended to be interesting and entertaining. I hope it will also make more people aware of the many ways birds act, here and in far places, how they have solved their problems and profited by their opportunities.

**STRAY FEATHERS FROM A BIRD MAN'S
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BIRDS USING TOOLS

[\[Ref\]](#)



MAN IS THE TOOL USER pre-eminent in the animal world, but he does not stand completely alone in this. Here and there, in quite different groups of animals such as insects, mammals, and birds, a few kinds have forged a little ahead of the rest of their near relatives and show the very beginning of tool using.

The song thrush of Europe is perhaps a borderline case. It feeds in part on snails. To get the soft edible animal out of its shell, it carries or drags the snail to a favorite rock, its anvil, and there hits it against the anvil until the shell is broken and its contents exposed. The question is, can this be considered as using a tool? If the song thrush moved or prepared the rock, which it does not do, there would be no question that it was a tool. The sea otter brings a stone from the bottom of the ocean and places it on its floating body to use as a similar anvil in cracking hard objects, and this undoubtedly is the use of a tool. At the other extreme are many species of birds that beat their prey on branch or ground, wherever they happen to be. The song thrush is certainly an advance over that, and can, I think, be considered as using a tool in a primitive way.

A few other species, too, bring shellfish to special places. Gulls on our coasts pick up mussels and clams and, flying over a rock or some other hard surface, drop the shellfish, and follow it down. If the shell is broken, the

dish is ready for the gull; if the shell is not broken the gull takes the shellfish up to a higher altitude and tries again. In places where hard-surfaced roads are conveniently located gulls have learned to use them as shell-breaking places, and such roads become littered with shells.

Crows of more than one species also use the same routine in breaking open shellfish, and they, too, have learned to use special hard surfaces, such as masonry walls, on which to drop the shellfish.

PAINTING A BOWER The satin bowerbird of Australia, a species known to science as *Ptilonorhynchus violaceus*, has also been considered as a case in point when discussing the use of tools. The birds are somewhat larger than a robin, the male glossy blue-black, the female greenish. The male of this species constructs an elaborate bower, presumably for courtship purposes. It makes it of sticks and twigs, and decorates it with bright and curious objects such as shells, feathers, bits of bone, and fruits, as do several other species of bowerbirds. But the satin bowerbird is unique in painting the inside of its bower. Fruit is crushed in its bill, and the bird, using its bill as the tool or paintbrush, smears the fruit juice on the sticks on the inside of the bower. While this is a wonderfully strange habit, and apparently unique in the bird world, it is doubtful if this behavior can be considered as using a tool. If the satin bowerbird used a twig, or a wad of moss or fiber, which it does not do, in spreading the paint, the case would be clear.

The clearest case is that of the woodpecker finch of the Galápagos Islands. *Camarhynchus pallidus* is its proper name. It is one of a group of dull-colored finches restricted to the Galápagos Islands. Before it became known that one species used a tool, the chief claim to fame of the group was that it, along with some other Galápagos Island animals, such as the giant tortoises, had a great influence on Darwin's thinking which resulted in his working out the theory of evolution as set forth in his *Origin of Species*.

The woodpecker finch feeds largely on insects it gets by searching and probing on the ground, and on the trunk and leaves of trees. In searching crevices the woodpecker finch is handicapped by its rather short, thick bill, and to offset this, it picks up a slender, short length of stick, or the spine of a prickly pear, and with this in its bill, pokes into crannies. The insects, disturbed or driven out, are seized. Sometimes the woodpecker finch digs

into the tree trunk and then gets a stick to probe with; sometimes it carries its probe about with it, poking in crannies until prey is disturbed. Then the stick is dropped and the food seized.

We have seen how several birds are perhaps borderline cases in using tools. They use certain special aspects of their environment in preparing their food, and use it time after time. It's probably instinctive behavior, but learning is shown in the gulls and crows coming to recognize and use a hard-surfaced road in breaking open their shellfish. The use of a probe by the woodpecker finch is a clear and unique case of tool using by a bird.

BIRDS AS BRIGANDS

[\[Ref\]](#)



ANTI-SOCIAL ACTIVITIES of humans such as those of brigands who plunder their fellow men find their parallels in the bird world.

The bald eagle is one of the best-known of the birds that practice such brigandage. Fond of fish, and capable of capturing it himself upon occasion, it is a common practice for the eagle to take fish from the osprey, plunder the osprey has just caught from the water. The osprey, with a fresh-caught fish, flies heavily. The watching eagle quickly overtakes the smaller, heavily laden bird and forces it to drop its catch, then dives down and usually catches the fish before it can strike the land or water. Rarely does the osprey escape with its food under such an attack. It is recorded that an eagle made several dives at one fish-laden osprey and, when these were not successful in making it lose its hold on the fish, the eagle dived under the smaller bird, turned over on its back, and with talons outstretched, snatched the fish from the grasp of the osprey and sailed away with it, as successful a pirate as ever sailed the seas.

NEMESIS OF VULTURES Besides taking fresh-caught food from the osprey the bald eagle has been seen pursuing vultures and making them disgorge their meal of carrion. The eagle, if unsuccessful in catching the disgorged food in the air, may land on the ground and eat it there. We know also that the aerial flights the eagle uses to frighten the vulture into relinquishing his food are

not idle threats, for an eagle has been seen to strike and kill a bird that refused to disgorge.

Not only does our American eagle adopt such practices, but related species in other parts of the world behave in similar ways. The New Guinea sea eagle harries the osprey there, and on the west coast of Africa a sea eagle robs pelicans and cormorants of their prey.

Certain long-winged birds of the tropical seas, such as *Fregata magnificens*, are known popularly as man-o'-war birds or frigate birds, reflecting their well-known character as pirates and tyrannical freebooters. The man-o'-war birds get part of their food from many creatures which swarm at the surface of the sea, but they also get much of their food by forcing terns, cormorants, boobies, and pelicans to deliver up their catch.

In a tropical bay a school of small fish comes to the surface, perhaps driven by large fish below; from far and near terns gather, darting down to seize the fish that jump into the air. Above them circle the frigate birds, ready to dive down and chase and harry a successful tern until it drops its fish and leaves its prey to the freebooter.

BOOBIES ARE VICTIMS Frigate birds may sail about, also, where a colony of nesting brown boobies is located, waiting for the birds laden with food to return home. When such a food-laden booby returns, the frigate bird dashes down at it, buffets it with its wings, snaps at it with its long, hooked bill, until the booby finally drops its fish for the man-o'-war bird to enjoy.

The skua, a big, dark relative of the gull, is also known as a pirate. Its chief food is fish but it also eats many other foods from the sea. It rarely takes the trouble to fish for itself but watches until some other bird, perhaps a gull or a tern, has been successful in its hunting and then gives chase, forcing the unfortunate hunter to relinquish its food. Several of the skua's smaller relatives, the jaegers, have similar habits. It is written of the pomarine jaeger off our New England coast that they are notorious pirates and freebooters, the highwaymen among birds that prey on their neighbors on the fishing grounds and make them stand and deliver. The jaeger gives chase to a tern that has caught a fish and follows it through every twist and turn as if the two were yoked together. Finally the harassed tern drops its fish and the jaeger swoops down and seizes it before it can strike the water.

BIRDS BATHING

[\[Ref\]](#)



THE TOILET of most birds includes wetting their feathers in water and shaking the feathers and preening them with the bill. This bathing probably helps remove foreign matter from the birds' plumage and helps keep it in good condition. In addition it is probable that in summer the birds derive enjoyment from the coolness resulting from the bathing. But birds bathe in cold weather as well as warm and have been recorded doing so when the temperature of the air was only 10 or so degrees above zero.

The sparrows and robins that come about a birdbath usually hop right into the water. They squat down, fluttering their wings, and duck their heads into the water, splashing and rolling it over their backs. They may become quite drenched. Then they fly to some perch to sit and preen and dry their soaked feathers.

But some birds take shower baths. During a shower in late summer I have seen marsh hawks sitting in the rain with wings spread, apparently enjoying the wetting the shower gives them, and a buzzard has been recorded as deliberately flying to an open perch in a rainstorm and sitting there with its wings spread and sometimes shaking them until the shower was over, when it flew to a sheltered place.

SPRINKLERS A BOON The artificial showers of lawn sprinklers provide an opportunity for birds about our gardens to take a shower bath in fine weather. A robin or a flicker may hop into the shower and squat there and indulge in bathing antics on the wet grass. Hummingbirds have been seen to fly into the dense spray of a lawn sprinkler and hover there for a moment, gradually assuming a vertical position and spreading the tail, then slowly settling to the ground, and finally "sitting" on the grass, body erect and tail spread out fanwise, the wings continuing to vibrate slowly. In a few moments the bird may rise into the air and repeat the whole performance.

In wet tropical forest it is probable that many of the treetop birds bathe in the water that collects on the surface of the leaves, pushing their way through clusters of wet leaves and over wet surfaces of others until they are as wet as if they had actually been bathing in water. This is not restricted to tropical birds, for even in our latitudes towhees have been recorded as bathing thus, and thrushes and flickers have been seen to rub themselves over the wet grass and then go through the actions of bathing followed by preening.

BATHING WHILE FLYING Watching swifts or swallows coursing low over the surface of a lake and occasionally touching it leaves one with the impression sometimes that the birds are bathing rather than picking up insect food or drinking. With some other birds the habit of bathing from the wing is more definite. Sometimes drongo shrikes that are sitting up on a perch near the edge of a pool will fly out over the water, drop directly into it with a little splash, and then rise and fly back to their perch, where they either repeat the performance or sit and preen their feathers.

POST-PRANDIAL ABLUTIONS Ospreys have been recorded as bathing while on the wing in a rather striking manner. They have been seen flying along just above the surface of the water, then descending into it, adopting a sort of vertical American-eagle attitude while flapping the wings, then rising a little, flying on, and repeating the process. It has been suggested that the osprey is washing its feet in this manner after finishing its meal. One observer makes this still more definite. He says that the osprey finishes its meal of fish on a perch in a tree and then flies low over the lake. Dropping

both its legs, the osprey drags them through the water, flapping its wings all the time. Then it immerses its beak and head into the water while still flying along, apparently washing off the scales and slime that it had gotten on itself while making its meal of fish.

HOW BIRDS ANOINT THEIR FEATHERS [\[Ref\]](#)



A BIRD'S PLUMAGE receives a great deal of care from the bird that wears it. The bill is the only implement for this grooming, and it is run through and along the feathers it can reach, helping clean them and making sure they lie in their proper place in the bird's dress. There are parts of the plumage that the bird's bill obviously can't reach, as that of the head, but ducks at least surmount this difficulty by rubbing their head against their body.

Many birds have oil glands (the only external glands that most birds have), a pair of glands just above and in front of the root of the tail, on the back. They contain an oily substance, and the usual explanation of its use is that the secretion of these glands is used in dressing the feathers. Certainly birds that have oil glands seem to use them, nibbling at them as though to press out the oil, touching them with the bill, and then rubbing the bill through the feathers, and rubbing the head against the oil gland.

The beautiful, soft, whitish bloom seen on some birds' feathers, such as the pale gray of a male marsh hawk and filmy appearance of some herons' plumage, is caused by specialized feathers called "powder down." Sometimes this powder down is scattered through the plumage; sometimes it is in patches, such as the particularly conspicuous ones in the herons. The tips of the powder down are continually breaking off and sifting over the rest of the plumage, giving it the bloom that with handling quickly rubs off.

WALNUTS AS A COSMETIC But birds sometimes rub foreign substances over their feathers—just why we don't know. Grackles have been known to use the acid juice of green walnuts in preening.

In Pennsylvania starlings have been seen to come to walnut trees when the nuts were almost three-quarters grown, in June, and peck a hole in the sticky hull of a nut, clip the bill into it, undoubtedly wetting the bill against the pulpy interior, and then thrust the bill into their plumage.

They did this from June to August, especially on hot, dry summer days, but some birds continued this even during light rain. Some years before the above was recorded, when this sort of thing was less known, Edward Howe Forbush, noted ornithologist, cautiously used a similar record in his classical *Birds of Massachusetts and Other New England States*. He writes that his colleague, J. N. Baskett, says he saw a bluejay lift its wing and rub pungent walnut leaves repeatedly into the feathers beneath.

BEER AND MOTH BALLS Since then such things have been recorded a number of times, including a catbird that anointed its feathers with a leaf and a grackle that found a moth ball and, holding this in its bill, rubbed it against the underside of its spread wing and the side of its body. After several applications the grackle dropped the moth ball and preened its feathers; then again it picked up the moth ball and treated the other wing as well as its belly.

Recent experiments with tame song sparrows have shown that they may use beer, orange juice, vinegar, and other things made available to them in dressing their plumage, and it appears that this may be correlated with a little-understood type of activity known as anting, in which live ants are placed on the feathers.

TRAVELING BIRDS' NESTS

[\[Ref\]](#)



IN SPRING and fall many of our birds make long journeys under their own power, some of the most publicized being the migration of the Arctic tern, a bird that may spend the northern summer north of the Arctic Circle and, before returning there next season, may have visited south of the Antarctic Circle. The golden plover that makes a nonstop flight to Hawaii is another famous traveler, and many of our smaller songbirds are no mean travelers either. The barn swallow that nests about an Illinois farm in the summer may spend the winter in Argentina. The tiny hummingbirds' feat of crossing the Gulf of Mexico nonstop is worthy of mention too. Such travels have become commonplace through familiarity. We have come to accept even the possibility of occasional transatlantic passages of small perching birds, helped by transatlantic vessels, and of such birds as starlings, making their way from place to place by boxcar.

But when it is time for birds to make their nests and rear their family we expect them to give up their traveling for a time and to settle down in one place. We expect, with our songbirds, to have the male arrive first, pick out a territory, and announce to his species that other males are to keep out and that a mate is welcome. The female arrives and chooses her mate or territory, and a nesting ensues. Many species defend the area around the nest against others of their kind. So it comes as a surprise to find nests built

in such a situation that they are not stationary but move back and forth, along with part of their environment.

BY BOAT Tree swallows nest on the ferryboats that ply between Ogdensburg, New York, and Prescott, Ontario, across the St. Lawrence River where it is more than a mile wide. The nests are tucked into suitable openings on the ferries, and the frequent trips back and forth across this mile of water and the docking at different piers do not seem to disturb the birds. They gather their nesting material of feathers and straws and leaves from either shore, and when the young are being fed, insects may be gathered about the Canadian or the United States shore, depending on where the ferryboat is docked.

Another example comes from Western Australia, also of a swallow, the welcome swallow which is nearly like our barn swallow. A pair of these birds nested on a boat used for visiting local coastal stations. If there were eggs or young in the nest when the boat sailed, the old birds would accompany it, once following her on a trip of thirty-five miles and back.

BY TRAIN Barn swallows have been noted nesting on railway trains that run across the two-mile portage between Atlin Lake and Lake Tagish in British Columbia. In the summer the train makes the trip almost daily, and for many years a pair, or a succession of pairs, has made its nest and raised its young in one of the open baggage cars. Members of the train crew took an interest in the birds and put up a cigar box for a safe place for their nest. Here the family seemed to prosper, undisturbed by the proximity of people and baggage and the clatter as well as the movement of the train.

MALADAPTATION IN BIRDS

[\[Ref\]](#)



THROUGH SELECTION birds have become adapted to their environment. In most cases this is successful adaptation. Occasionally, however, we come across instances in which the adaptations do not work out. Such cases, where the actions of the birds are not beneficial or are even detrimental to it, come as surprises.

Since the introduction of the Tartarian honeysuckle (*Lonicera tatarica*) into the United States from Asia, its planting as an ornamental shrub provides each autumn a display of juicy red fruit. This fruit contains saponin, a substance that has the effect of an anesthetic and muscle poison and may paralyze the greater nerve centers (in sufficiently large doses saponin causes death by cardiac paralysis). A condition of intoxication has been recorded for robins feeding extensively on these honeysuckle berries: "... this drunkenness has been seen in every shade of severity, from mild unsteadiness to a degree of incoordination sufficient to cause the birds to fall to the ground. It seems to make some of the birds utterly fearless and perhaps a bit belligerent, for they become quite unafraid of passers-by and interested spectators. A few dead robins have been found about these honeysuckle bushes—presumably poisoned by the berry diet." Fortunately the poisoning of birds by this honeysuckle seems to be uncommon.

In the Philippines the local people gather the juice of the coconut inflorescence in bamboo tubes placed in the crowns of the palms. This juice

ferments quickly and provides a refreshing, mildly intoxicating drink. A little parrot of the Philippines, the hanging parakeet, has a taste for this drink, comes and drinks from the containers, sometimes becomes drunk, falls in, and drowns.

The California woodpecker ordinarily differs from many birds because it does not lead a hand-to-mouth existence but stores food. These woodpeckers feed extensively on acorns, and one way they store them is by drilling holes in the bark of a tree and fitting an acorn into each hole. The whole trunk of a tree thus may be pitted with stored acorns. When the acorn crop fails and the nuts are scarce the woodpecker goes through the same storage activities but, being unable to find sufficient acorns, it stores pebbles instead. These pebbles are, of course, quite useless to the woodpecker, and this is an interesting example of an instinct "gone wrong."

Sometimes these woodpeckers have another method of storing their acorns. This is by dropping them into cavities in tree trunks, but when stored in such a way there seems to be no way for the birds to reach them. Here again we have a blind impulse to store acting in such a way that the bird gains nothing by the act.

The raven is ordinarily and quite correctly considered one of the most intelligent of birds, but a raven I kept in captivity and fed small fish attempted to store some of them by pushing them through a knothole in the back of its cage. The fish fell about fifteen inches below the knothole, where the raven could not possibly reach them. After pushing each fish through the raven peered through the knothole though it could not see the fish. Here again we have the instinctive storing act carried out in such a way that it produced no benefit to the bird.

The late George Latimer Bates, noted ornithologist, studying the birds of West Africa, found a most surprising thing in connection with one of the honey-guides. As a group, these birds are noted for the habit of attracting the attention of human beings and leading them to bee trees, presumably so that they will break down the bee tree for the honey, and the birds can feed on the scraps left over. Bates found that the West African species is parasitic on other birds in its nesting habits and its young have been found in the nesting hole of a little barbet. This barbet was a much smaller bird than the honey-guide and the entrance to the nest hole was so small that Bates

doubted that the honey-guide would have been able to get in to lay its egg. He suggested that the egg may have been laid elsewhere and deposited in the nest by the parent's bill. It is difficult to understand how the young honey-guide would be able to get out, for when fully fledged it would have been far too large to squeeze through the entrance that admitted the tiny body of its foster parents, the barbets. This is an almost incredible story and if true looks like a case of maladaptation.

FEATHERED BABY SITTERS AND CO-OP NURSERY NESTS

[\[Ref\]](#)



CO-OPERATIVE NURSERIES, where a few parents look after the young while the rest of the adults, temporarily freed of the care of their offspring, can go about their other affairs, appear in the bird world.

The wild turkey of our Eastern United States commonly steals away singly to lay its eggs and incubate them in its nest on the ground. But occasionally it happens, Audubon writes, that several hen turkeys associate together and lay their eggs in one nest, and raise their young together. With the turkey apparently there is little division of labor, as Audubon writes of finding three hens sitting on forty-two eggs, but he says that one of the hens is always on the watch at the nest so that natural enemies have no chance to rob it.

A GREGARIOUS BIRD What is of only occasional occurrence in one species may be the regular course of events in another, and in the ani we find it customary for a number of birds to nest together. The anis are moderate-sized cuckoos living in the tropical Americas. The smooth-billed ani is perhaps the best known, for Dr. D. E. Davis, when studying at Harvard for his doctor's degree, made a special trip to Cuba to study them in the field. The smooth-billed ani goes in flocks the year round. Usually there are about

seven birds in the flock, but there may be as many as twenty-four. The nest is a bulky structure of twigs and fresh leaves. When nest building starts usually one bird is most active, but as many as five birds were seen carrying in sticks at one time. When the nest of sticks and leaves is finished several females may lay their eggs in it. But apparently only one bird incubates at a time, and the male takes his turn at incubating. When the young hatch, after about thirteen days, most of the adults in the colony help feed the young.

Eider ducks may nest in dense colonies, but each bird has its own nest in which it lays its own eggs, and in which the female alone incubates. But after the young hatch and the mother leads them to the water, the young may band into larger flocks, accompanied by a number of females, and the young seem to be independent of their particular parent, but attach themselves to and are tended by the nearest duck.

PENGUIN SOCIAL GROUPS A much more elaborate system for caring for the young has been evolved by certain penguins. The sexes alternate in their care of the young in the early stages. But when the young are partly grown the family unity breaks up for a communistic type of social organization. The young are now grouped into bands of up to twenty or more birds and are left under the care of a few old birds, while the rest of the adults go to the water, which may be some distance away. Periodically they return with food for the young. Apparently the individual young is not recognized by the parent, which goes to the particular group of which its young is a part, and there may feed any one of the "child groups."

Here we have two definite cases of a social organization that has resulted in division of labor: in the incubation of the ani, and in the care of young penguins. In addition we have two less specialized cases of the same thing, showing the sort of raw material on which evolution can operate to produce new behavior patterns.

BIRDS' NESTS AND THEIR SOUP

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IN CAVES near the ocean in the Far East nest myriads of tiny swiftlets whose chief impact on the civilized world is that their nests provide an edible article of commerce. "Birds' nest soup" at once comes to the mind of the Occidental, few of whom have ever eaten of the nests, or even seen the birds to know them. For those who would like to see the nests, some museums have them on exhibition, such as in the Chicago Natural History Museum, where two nests are placed in their natural setting, and beside them is a quantity of the material of commerce in its raw state.

The birds themselves are dusky-colored swifts only a few inches long, and belong to a group of swifts that represents perhaps the most puzzling problems of species identification in the bird world. As yet we do not know even how many species there are. The genus is called *Collocalia*. Only some of its members make the edible nests; others mix so much moss into the nest that it is useless for soup. One species has the scientific name of *esculenta*, given in reference to the supposed edible nature of the nest, but through error the name was applied to a species whose nests are not edible. In habits all these swiftlets seem very similar, flying about with a rather weak flight for a swift, catching their insect food on the wing.

A number of swifts, including our chimney swift, use the secretion of their salivary glands as a glue to stick together their nest, and to stick it to the

wall of a cave, the inside of a hollow tree, or the inside of a chimney. But some of the edible-nest swifts go further and make their nest entirely of this secretion from their enormously enlarged salivary glands. This material, as it comes from the mouth of the bird, resembles a saturated solution of gum arabic and is very viscid. If one draws out a strand from the mouth of the bird and sticks it on a rod, by rotating the rod and winding up on it the thread of saliva one can empty the salivary glands of the bird. This material dries quickly, and is the material of which the nest is made. When the bird makes its nest, which it does in large colonies in caves, it flies up to the rock wall, applies the saliva to the rock in a semicircle or horseshoe. Gradually a little shelf is built out, and in the finished nest one can see the many little strands that have gone into the structure. It may take the birds as long as three months to make this nest, even if undisturbed. The birds lay their two eggs in the nest, and raise their naked, helpless hatchlings into facsimiles of themselves in it.

But in the Orient, especially in China, the nests are highly prized by epicures as a delicacy. As the supply is limited the price is high. A note with some material we saw stated that the price was \$12 to \$36 a pound in Siam.

The climbing for and collecting of these nests requires daring, skill and is not without danger. The nests may be far back and high up in the cave. Ropes and poles may have to be fixed in place to aid the climber, who has a flaming torch in one hand and carries a sack or basket for the nests. In Siam, at least, the collecting of these nests was hereditary, father training son. The rights to collect nests are valuable. In Siam, where the rights to collecting the nests were vested in the state, revenue of as high as £20,000 has been received from the rights for this collection.

The nests are said to be of highly nitrogenous material, and contain about 50 per cent of protein and 7½ per cent of mineral matter. Their use as food is an Oriental custom, but an Occidental opinion of their flavor is that it is bland, and an appreciation of it needs to be cultivated. The price of these nests is so great that unscrupulous persons have manufactured spurious nests. These nests are made from agar-agar, the jelly made by boiling down certain seaweed, and are so cleverly flavored that only connoisseurs can detect the fraud.

We usually think of these nests in connection with birds'-nest soup, which may be made with chicken or beef broth and then the cleaned material of the nest added like tapioca or vermicelli. Sometimes a sweet soup is made. Sometimes lotus seeds, sugar, and the nest material are used in the preparation of the dish. But in the Orient, at least formerly, they're considered to have medicinal qualities, too. It is said that when combined with ginseng they are capable of restoring life to a person on the point of death. In Northern China where the winter is bitterly cold, it is a general belief that the blood congeals and can only be thawed out by drinking a soup made of these nests. The list of further benefits, such as against tuberculosis, as a tonic, stimulant, and a pacifier of the stomach, recall advertisements of patent medicines.

WALLED WIVES OF HORNBILLS

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FOR LONG it has been written that the male hornbill walled up his mate in her nest in a hole in a tree at nesting time, and one author even wrote that the male plucked out the female's feathers at this time. The facts underlying these statements have different interpretations, but the nesting of the hornbill is still one of the most extraordinary of animal habits. Travelers and naturalists in Africa had brought back tantalizing bits of information, to add piecemeal to our knowledge of these birds. Now all this is synthesized and corrected by R. E. Moreau, onetime resident in East Africa, who made a study of certain species, raised young birds by hand, and gave us a comparative study of their behavior. Even this study must be considered preliminary, for, of the twenty-six African species, we have breeding data on only sixteen of them.

First we must not generalize too far as to "the hornbills," for there are Asiatic and Malayan species as well as African, and African species differ among themselves, the ground hornbill being especially aberrant in its habits.

It is quite true that in many African species the female is walled up in her nest, and the period when she is enclosed may last three to four months. But it cannot be interpreted as an imprisonment forced on her by the male, and presumably she could, if she wanted to, open the entrance at any time, as she does finally on emerging.

Among the African species the details vary, but the nest is usually located in a hole in a tree, and except in the case of the ground hornbill the entrance is plastered up so that only a narrow slit is left, about wide enough for the passage of the bird's bill. The female takes an active part in the walling up of the opening, and might be said to wall herself in. When the opening to be filled in is wide, the male may bring earth, which he mixes with saliva in his gullet, and presents to the female, who does the actual plastering. In some species the walling up of the entrance may take months.

The female may wall herself in some days before she lays her first egg. Throughout incubation she remains there. Depending on the species, she may peck her way out, or burst out when the young are partly grown, or she may stay until the young are ready to fly.

During the time the female is walled in the male brings food for her, and later for the young, also. That he is a good provider is indicated by the fatness of the female and her young. This is proverbial with the natives of Africa. The method of feeding varies with the species. The male may bring a bit of food in its bill, pass it in to the female, and then go for another, or in other species we might think more intelligent, the male carries a quantity of berries in its gullet, and these are regurgitated one by one and passed to the waiting female; such species make trips to the nest less frequently.

Apparently shortly after the female goes into the retirement of her walled-in nest, she molts all her flight feathers, so that she is flightless, and then begins to grow them again.

When the female bursts out of the nest with the young only partly grown, the young that remain in a still very undeveloped state in the nest, using material in the nest such as remains of food and rotten wood, replaster the hole! The young, perhaps only halfway through their fledgling period, wall themselves in! The female then helps the male care for the young.

Such is an outline of what some of the African hornbills do at nesting time. The habit is unique in the bird world. One species appears not to wall up its nest. In an Asiatic species it is said that if the male is killed other hornbills help to feed the female in retirement. The whole procedure is an amazing behavior pattern, and one for the development of which it is difficult to find a functional explanation.

BURIED EGGS AND YOUNG

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THE CROCODILE BIRD, or Egyptian plover, has enjoyed a dubious publicity because of its reputed habit of entering, and coming out of, crocodile mouths. As Herodotus put it, the crocodile's mouth is infested with leeches, and when the crocodile comes out of the water it lies with its mouth open facing the western breeze. Then the crocodile bird goes into the crocodile's mouth and devours the leeches, to the gratification of the crocodile, who is careful not to harm the bird. Though there are some more recent observations corroborating this, modern observers who have had abundant opportunity have watched for this behavior and have not seen it.

As one authority on African birds puts it, it is evidently not an everyday occurrence.

But the crocodile bird has other habits that are just as bizarre and interesting. It lives along the sandy shores of African rivers, and when it lays its clutch of two to four eggs these are buried in the sand so there is no sign of them aboveground. The bird sits on top of this spot. A. L. Butler, who studied this bird in the Sudan, thought that the sand might be scraped away from the eggs and the eggs brooded in normal fashion by night. The young birds are very precocial, and feed themselves on tiny insects, but they follow the parent. When danger threatens the young squat motionless in some depression. The toe mark of a hippopotamus is a favorite place. Then the old bird, with her bill, throws sand over the young until they may be completely covered. Not only does this happen when the birds are very

small, but continues up until the time the birds can fly. Dr. W. Serle in Sierra Leone once saw a crocodile bird burying something and found the disturbed spot fairly easily, as recent rain had beaten the sand beach smooth and hard; a fully fledged young was unearthed. It squatted motionless until prodded from behind, then it ran swiftly, rose, and flew away strongly.

The burying is not only protection from immediate enemies; A. L. Butler believed it was normal for the young when not feeding to be buried for safety or as protection from the burning sun. For a further protection from the sun the parent moistens the sand by regurgitating water over it.

Butler on one occasion saw a crocodile bird drink at the water's edge, run up onto a sand beach, regurgitate water, then settle to brood. Butler marked the spot, went to it, and, scraping away the dampened sand, found a tiny chick about one inch below the surface.

This covering of the eggs by the parent is not unique in the bird world. The pied-billed grebe of North America also does this. When disturbed at the nest the incubating bird has been seen to use quick pecking motions to draw material from the edge of the nest over the eggs. Instead of leaving the eggs exposed the nest simply looks like a heap of trash and may thus escape the attention of a predator. It used to be thought that this grebe used to incubate only at night, leaving the eggs covered during the day to be incubated by the heat from the sun and from the decaying vegetation of the nest. However, recent studies have shown this is not the case, and protection by concealment seems to be the main advantage of this behavior.

Yet another species of quite a different group, the eider duck, covers its eggs on leaving them. The eider's nest is characterized by a blanket of down, plucked from the breast of the bird, and when the female has time, when she leaves the nest she pulls the edges of the down blanket over the eggs, perhaps for concealment, perhaps for the sake of the down's insulating properties, keeping the eggs warm in a northern climate during the parent's absence.

Here we have covering of eggs for what seems to be very different purposes: to keep the eggs cool; to keep them warm; and to hide them from view.

THE SNOWY OWL AS A TRADE INDEX [\[Ref\]](#)



ANGUS GAVIN was a fur trader at the Perry River post of the Hudson's Bay Company on the edge of the Arctic Ocean. White foxes were the chief fur brought in, and the Eskimos were the trappers. Sometimes it was necessary to advance credit to an Eskimo, against the expectation of a coming season's catch out of which the advance was to be repaid. Gavin, who was a keen naturalist as well as trader, writes, "I used my observation on Snowy Owl abundance to govern extension of credit...." When snowy owls were abundant he could extend liberal credit to the Eskimo with every assurance the white-fox catch would be good and that the Eskimo would be able to liquidate his debt. When snowy owls were scarce little credit would be extended, for the white-fox catch would be small.

In general we've accepted the value of birds to man, and are appreciative of the complicated web of life in which one animal affects many others. But this use of snowy-owl abundance as a guide in granting credit strikes me as novel. Actually, of course, it is quite sound, for it uses one part of the chain that links such diverse items as owls, lemmings, foxes, Eskimo, fur trader, and finally of course milady in her white-fox furs.

LEMMINGS IMPORTANT First of the factors involved is, of course, the vegetation; the grasses, herbs, and tiny dwarf shrubs of the Arctic barrens. The next are the lemmings, mouselike creatures of the Far North that eat the

vegetation. They are the first step in turning grass into flesh and fur and feathers. One of the striking facts of lemming biology is the fluctuation in their numbers. Some years they swarm, lemmings are everywhere, and in places they erupt in vast emigration, the tundra and the sea ice being covered with masses of moving lemmings. We know this best from the accounts written about the lemmings of Norway, but the same thing occurs in the American Arctic. At other times they're scarce and it is difficult to find even one. Strangely there's a periodicity in this, and periods of abundance and scarcity tend to recur every four years. What happens or what causes it we don't know.

The Arctic fox, staple fur bearer of the Far North, and the snowy owl both prey on lemmings. Lemmings are so important to them that when lemmings are abundant the foxes and the owls prosper and multiply; when the lemmings are scarce the foxes and the owls starve or migrate, in any case where there are few lemmings there are few foxes or owls.

Thus we see how it is that an abundance of snowy owls can indicate that the Eskimo will make a good fox catch and the trader will do good business.

MONKEY BIRDS

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BIRDS GET their everyday names in a variety of ways in the countries where they live; from their looks, like the snake birds and the pond scroggins; from their color, like the cardinal and the blackbird; from their behavior, like the frigate bird and the creepers and the boobies and king-birds; from what they eat, or are supposed to eat, like the antbirds and plantain eaters and bee eaters; from what they say, like the poor-will and the more-pork; from how they say it, like the warblers and the screamers; from how often they say it, like the brain-fever bird and the wideawake terns; from where they nest, like the cliff swallow and the house martin and the chimney swift; and some from their non-bird associates, like the cowbird, moose-bird, and the monkey bird.

It is the monkey birds that have taken our fancy at the moment. The forests of Africa, the jungles of Borneo, and the forests of the Philippine Islands each have a bird that associates so often with monkeys that this habit became incorporated into its local name. The birds are not at all closely related. One is a hornbill, one is a drongo shrike, and one is the fairy bluebird. The hornbill goes in parties of their own kind, but apparently the drongo, and certainly the fairy bluebird prefer the society of monkeys to that of their own kind.

The stories we have of them stress the utilitarian aspect of the association; that the monkeys as they travel about through the trees scare insects out of their hiding places and the birds, being on hand, can snap up the insects more easily than if they had to search them out for themselves.

The monkey bird in Africa, which is a hornbill, follows, along below the monkeys in the lower branches of the trees. It used to be thought this was for the fruit the monkeys dropped, but then it was found the hornbills were insectivorous. Instead of being scavengers the hornbills are using the monkeys to beat out their game for them.

Hamba Kerah, the slave of the monkeys, is what the Malays of Borneo call the racket-tailed drongo. This is from its habit of stationing itself behind a band of monkeys traveling through the forest. But Mr. Ridley, who watched them, decided it was the other way around; the monkeys, unwittingly of course, were working for the drongo, acting as beaters to drive out the insects which the bird snapped up in the air.

In the Philippines it is "the sentinel of the monkey" that is applied to the fairy bluebird. The bluebird seldom associates with its own kind, but is almost invariably associated with a band of crab-eating macaques. But here again it seems the monkeys are acting as beaters for the bird, driving out insects.

This is a sort of unconscious co-operation one finds in the bird world. One animal helps out another without being aware of it. Birds are ever ready to profit by such behavior, and when it proves of enough benefit, the habit can become usual for the species, as in the cowbird-cow relationship, or indispensable as with the oxpecker-hoofed-animal association.

BIRD-MADE INCUBATORS

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INCUBATORS as we know them on chicken farms are electrical gadgets with thermostats to control the temperature, or at least with oil lamps to supply the heat necessary for the young chick in the egg to grow. Naturally we wouldn't expect anything so artificial as this in the bird world, but there is one group of birds that does not brood its eggs but has employed another method of incubating.

The birds that do this are fowl-like birds of the Australasian area. They are variously called "mound builders" from the nest mound they construct, "megapodes" from the large size of their scratching feet, or bush turkeys, presumably from their edible qualities. These birds bury their eggs and leave them thus to hatch without any brooding by the bird. The birds have adapted their habits to two different sources of natural heat. On some of the Pacific islands there is local volcanism making the sand warm. To such places the birds come to bury their eggs.

But in many of the tropical forests there is not this convenient natural heat. Another method is employed. The birds take advantage of the heat generated by rotting vegetation. They scratch up the surface litter of the forest floor into mounds—structures that may be a yard or two high and five or six yards across. Some much larger have been observed. It is into these the hens burrow and lay their eggs. The temperatures in them have been recorded as 95° to 96° F., which compares with normal bird

temperatures of just over 100° (bird temperatures are a few degrees higher than normal human temperature).

The bush turkeys from Queensland have been bred in captivity, and have given some extremely interesting data, according to an article by Mr. Coles in the proceedings of the Zoological Society of London for 1937. It was the male who did all the building of the mound. Though the female started to cover the eggs laid singly in burrows in the mound, the male finished this. And it was the male that looked after the nest mound during the incubation period, continually scratching over the surface layer. Both parents helped the young emerge, by digging burrows into the mound which the emerging young, who had started to burrow out, could use.

The young are in a very advanced state and apparently are able to fly and look after themselves upon emerging. On the day after hatching one chick is reported as able to flutter up to a perch six feet high. In the captive birds mentioned above, the parents, though they were attending to the mound and helped the chick out, appeared to take no further interest in the chick once it was out.

There are a few other cases when birds cover or bury their eggs. With the grebes it has been said they covered them and left them to be incubated, but that is doubtful. Certainly the megapodes are the only ones to present a clear case of "artificial" incubation.

This burying of eggs by the megapodes of course brings to mind the way some reptiles, such as turtles, bury their eggs. And considering that from an evolutionary viewpoint birds are really only modified reptiles, it is perhaps not surprising that they too have this habit. But that it is really an ancestral trait retained by the megapodes is doubtful. Rather I'm inclined to think it's another example of the many ways birds have evolved, or changed their habits so as to utilize as much of the environment as they can in as many ways as possible.

CORMORANT FISHING

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IN WESTERN EUROPE, when falconry was in favor for taking game on land and in the air, there was a certain vogue for training cormorants to take fish. Like the falcons, the cormorants were hooded and carried on the wrist, but of course where the falcons flew to their game, the cormorants swam to theirs.

It was in China where cormorants were domesticated, "completely and perfectly," as that eminent Sinologist Dr. Laufer says. Extensive breeding establishments have been maintained. The eggs of the breeding flock of cormorants are given to a hen to hatch, for cormorants as mothers prove unsatisfactory under domestication. When the eggs hatch the young cormorants must have special care; for, unlike the young of ducks and geese, young cormorants at hatching are not down-covered and able to run about, but are weak, helpless things sensitive to cold. They are placed in cotton batting, artificial heat provided when necessary, and they are fed by hand on a diet composed basically of chopped eel.

Finally the young are full grown and fully feathered. The training is now started. First the young are tied to a stake at the water's edge. A whistle signal is given and the young cormorant is pushed into the water. Thus he learns to know and obey the signal to go into the water. Then the trainer throws him little fish. These the cormorant catches in its beak and when he does the trainer whistles another signal, to bring the bird back to him with

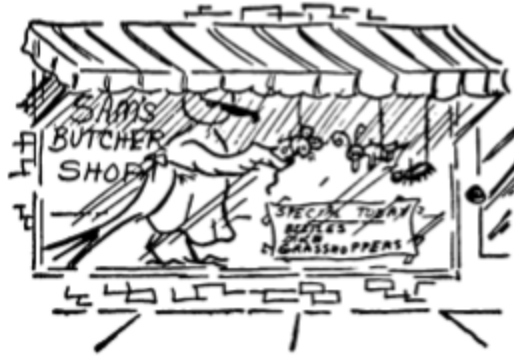
the fish. And the cord tied to the bird is used to demonstrate what is meant and make sure its done. So the training goes on until the bird has graduated to a class taught from a boat. Sometimes a small float is attached to the cormorant by a short cord, and it can be drawn in with a bamboo hook. If young birds are trained in the company of trained birds, it takes but half as long. Finally the training is complete and the fisherman sets out with his birds. This is no sporting event; it is the serious business of life, getting a living from fishing. On the sampan or the bamboo raft there may be from two to a dozen birds; sometimes they may have special perches built for them along each side of the boat. Sometimes the cormorant has a cord or band around its neck. The reason for this is disputed. Some say its a place to attach a cord; a place to get hold of the bird; some say each man's cormorant is thus specially marked for identification; some that it's to prevent the bird from swallowing its prey. With well-trained cormorants it is sometimes dispensed with. At a signal the cormorants go into the water, swim, and dive seeking fish. The fisherman, by stamping his feet, by voice or whistle, and by hitting the water with a bamboo directs and encourages the birds. When the cormorant catches a fish it brings it back to the boat, and the fisherman may use a net, or may lift up the cormorant onto the boat on an oar or pole, and take the fish from the bird. If a bird is lazy it's encouraged by beating the water near it with a bamboo pole. As cormorants' plumage is only partly waterproof they cannot stay in the water indefinitely, and this, as well as fatigue, probably determines the rest periods when the birds are lifted aboard. Sometimes the fisherman helps attract fish to the boat for the cormorants to catch by scattering grains of rice in the water.

When the day is finished the cormorants are collected, fed, and the fisherman goes home with the sustenance for his family, gathered by a bird.

In Japan the cormorant is also used, but apparently somewhat differently. There cormorant fishing may partake of the nature of a sport. Sometimes the cormorants are "harnessed" into a team, each attached by a cord to a single line, directed by one master. In China the fishing is usually done during the day, but in Japan night fishing is common, the scene being illuminated by fires in braziers or cressets on the boat, or lanterns.

THE SHRIKE'S LARDER

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OUR NORTHERN SHRIKE is a songbird which has developed feeding habits along the lines of those of a hawk. Whereas most birds its size are content with fruits, seeds, or insects of a size it can beat or bite and then swallow whole, our northern shrike takes not only small insects but prefers large ones, and mice and birds too big to be swallowed whole. It is an opportunist and takes what is most abundant and easily accessible. The shrike's strong hooked bill is a powerful weapon, used with a nipping motion that is directed at the back of the head or neck of mouse or bird.

Now with the dead sparrow or mouse the shrike is at a disadvantage. With a powerful bill hooked at the tip its feet are still those of a songbird and are not strong enough to hold its large prey while pulling it to pieces. Only small insects are held in one foot and pulled to pieces. To meet this need for holding large dead prey the impaling habit was evolved. The result of this is the so-called larders, which form a fancied resemblance to meat hanging in a butcher shop, and have given the birds their name of butcherbird. A thorn tree, a splintered end of a branch, or even the barbs of wire fencing may serve. The shrike flies to one of these, carrying the prey in its bill (rarely in its feet), and with a pulling motion fixes the prey on a projection point. Sometimes instead of impaling the mouse or bird it pulls it into the fork of a branch, and so wedges it there. Now the food is firmly held, and the shrike can use its bill effectively to pull off pieces of flesh and swallow them. When the bird has fed, it leaves the rest of the animal hanging where it was.

It may return to this food and make repeated meals of it if not spoiled, or dried up, until the whole is devoured. But often parts of meals are left hanging and discarded. If suitable thorn bushes are scarce the shrike may return time after time to the same tree with its prey, and in time this tree may come to be decked with many partly devoured carcasses. Such trees are the so-called "larders." There is another aspect of shrike behavior that adds to these larders. The shrike, even when replete, may seize any prey that appears and impale it. The bird's organization is such that the sight of a small moving animal may start the actions that end with impalement even when the bird is not hungry. This food usually is not eaten later.

Thus the shrike's "butcher shop" is not primarily a store of food, even though it sometimes serves as such when in times of scarcity remains of old meals are eaten. It is not a gathering of food in time of plenty and saving it for a later use. Rather the placing of many items in one tree is the result of its being a favorable impaling place. And the impaling is behavior developed to overcome the weakness of the claws in a bird whose disposition and strong beak enable it to prey habitually on larger animals which otherwise it could not tear to pieces and eat.

BIRD FLAVORS

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PARTICULARLY in the study of insects it has been shown that bright or contrasting and conspicuous colors tend to be associated with ill-flavor in the animals that wear them, while insects with a good flavor tend to be so colored that they are difficult to see. The first is a warning coloration—advertising to a predator that he will not enjoy eating this insect and better leave it alone; the other is concealing color, its function apparently to keep predators from finding their prey. The tasters in the experiments that have been used to work out the above generalizations were usually birds, but, as checks, a variety of other animals were used, and the magpie moth (*Abraxas grossulariata*), for instance, was found to be distasteful to certain spiders, frogs, lizards, various birds, a bat, and finally "the late Dr. Hans Gadow (one of the leading ornithologists of his day), who made a practice of sampling caterpillars, remarked on trying an *A. grossulariata* that it was quite one of the worst he had ever eaten!" Apparently ideas in taste are similar throughout large sections of the predatory animal world. Reversing the usual role, and using insects (hornets) as tasters of bird flesh, the celebrated British naturalist, Dr. H. B. Cott, has recently studied the question of the palatability in birds and their coloration. Naturally Dr. Cott, with his customary thoroughness, compared hornets as tasters with other animals, including cats and men, and found a surprisingly close agreement in the results.

The experimental procedure was to expose the flesh of two different birds (without feathers) at the same time, and see which the wasps ate first. Thus a graded series was built up of the 38 species of birds tested, with a palatability rating of from 1 to 38. The wryneck and the crested lark stood at the top of the list, and the pied kingfisher and the white-rumped black chat, as the least palatable, at the bottom with Numbers 37 and 38.

Then, surveying the coloration of the birds, and their habits, Dr. Cott made the important correlation that in general the birds whose flesh was most edible were protectively colored, and those whose flesh was least palatable tended to be conspicuous in color and behavior!

To relate it to the theory of evolution Cott concludes that selective pressure by predators seems to have forced vulnerable species along two divergent lines of specialization: leading in those which are relatively palatable toward concealment, and in those which are relatively distasteful toward advertisement.

HOW MANY FEATHERS HAS A BIRD?

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THE QUESTION as to the number of feathers on a bird seems a simple one without complication. Dr. Wetmore, the well-known ornithologist who was secretary of the Smithsonian Institute, has given us some data. The number varies with the species, of course: the smallest bird, a hummingbird from Cuba, had the fewest, 940 feathers; larger birds had more, the robin 2587, and the mourning dove 2635 feathers. A glaucous-winged gull had 6540; a mallard 11,903 feathers; a Plymouth Rock chicken was said to have 8325 feathers; and a later investigator reported 25,216 feathers on a swan.

But as one thinks of it, more questions arise, as in any investigation. The answer to one question poses two more. The first question is, do not the birds in winter need a wanner plumage to keep out the cold than they do in summer, when it is warm? Do they have more feathers then? This was definitely true in the case of the goldfinch: a bird in summer dress had only 1439 feathers, while one in winter plumage had 2368 feathers, obviously an adaptation for cold weather.

The next question is more abstruse, but eminently practical: the smaller a body, the larger exposed surface for its weight it presents. That is, for its weight a small bird has a proportionately much greater surface from which heat is lost than does a larger one. With equal heat-producing mechanism and metabolism, a small bird would need more insulation than a large one. Reduced to its simplest: one would expect small birds to have relatively

more feathers than large ones: more feathers per gram of weight. Is this true? Two members of the Department of Poultry Husbandry at Cornell University, Dr. F. B. Hutt and Lelah Ball, supplied the answer. Small birds do have more feathers per gram of body weight than do larger ones. A hummingbird weighing 2.8 grams had 940 feathers or 335 feathers per gram; a nighthawk weighing 67.9 grams had 2034 feathers or 29 feathers per gram; while a swan weighing 6123 grams had 25,216 feathers or 4 feathers per gram of body weight.

Presumably there are still other relations: Do the birds that live in the tropics where it is warm have fewer feathers than species of the same size of arctic climates, as one would expect? Are certain types of feathers such as those of aquatic birds better insulated than those of land birds, so that the bird requires fewer of them to keep warm? Does a dense coat of down reduce the number of feathers needed to keep warm? Do the loose feathers of ostriches, lacking barbules, necessitate some adjustment in numbers? The things we've learned point the way to other questions to be investigated.

LAST YEAR'S BIRDS' NESTS

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THE WISDOM of our fathers is sometimes embodied in what we call old saws, to wit, "Many hands make light work," to which the iconoclast retorts, "Too many cooks spoil the broth." And when we come to the phrase, "As useless as a last year's bird's nest," we must reply, "Circumstances alter cases." For many a bird's nest of yesteryear still has its use; some a biological use to other birds; some to feed and clothe man.

SUBLEASES The snug, secure cavity that a woodpecker chisels in some tree trunk for its nest will last for many years, a shelter in which tree swallows, house wrens, screech owls, bluebirds, or wood mice may make headquarters and use as a nursery. In the strange forests of saguaro, a giant cactus of southern Arizona, the nest cavities of the gila woodpeckers and the gilded flickers in the cactus trunks seem necessary for the presence of many nesting birds. Without them the birds would have to go elsewhere for cavities in which to nest. In old woodpecker nest cavities the elf owl, pigmy owl, screech owl, sparrow hawk, ash-throated flycatcher, martin, and crested flycatcher commonly nest, and cactus wrens and even Lucy's warbler may use them. Their use is not confined to birds alone, for scaly lizards, snakes, rats, and mice have been found in them. In the Argentine there is a woodhewer that appears to depend on the domed mud nest of the red oven-bird for its nesting sites. It takes over a recently vacated or an old nest of the oven-bird and lines it with grass and feathers for its own use. In

Africa and Madagascar the great domed nest of the hammerkop stork may find a secondary use in sheltering barn owls.

SANDPIPERS AND EAGLES But it is not only burrows and domed nests that when deserted by their original occupants are used by other birds. The solitary sandpiper of our northlands belongs to a group in which nest building is reduced to a minimum, usually little more than a hollow in the ground with a few bits of material added. But the solitary sandpiper, and the green sandpiper of the Old World have broken with tradition and customarily lay their eggs in the abandoned nest of some thrush. Our great homed owl is another bird that may use the discarded nest of a crow or hawk for its eggs and young. And age in the eagle's nest means little to the eagle. Frances Herrick, the noted chronicler of the life of the American bald eagle, writes of one nest in the crotch of a lofty tree that had been in use for thirty-six years. Each year more material was added until the nest became 12 feet high, 8½ feet across the top, and was estimated to weigh 2 tons.

Man has found, among others, the following two direct uses for two kinds of birds' nests: one he uses for food; of another he makes covering for himself.

The swift's nests used for food have been discussed in another chapter, "Birds' Nests and Their Soup," so here I will only tell of the use of birds' nests as human covering.

EIDER-DOWN BLANKETS An eider-down has come to mean a comforter, a sleeping bag, or even a padded jacket. But to an ornithologist eider down still has its older meaning: the down of an eider duck. It is this material gathered from the eider ducks' nests which forms the article of commerce. The eider's nest may contain grass, seaweed, and sticks, but it is notable for the blanket of down on which the eggs rest, and with which the female covers the eggs when she leaves them. This down is plucked from the breast of the female. If it is taken from the nest she replaces it with more, and it is on this principle that harvesting of the down is carried out. On islands and islets in the northern part of the North Atlantic eiders nest in great numbers in dense colonies. Some of these are jealously guarded by the local inhabitants, who gather the first blanket of down from the eggs, and later,

after the eggs have hatched, gather the second crop of down with which the female has replaced the first to guard her eggs against the inclement weather of those boreal latitudes. Each nest may yield an ounce or so of the precious down, which is carefully cleaned and sent to market. It is this material, extremely light, extremely elastic, and one of the best non-conductors of heat, which finally becomes the important part of real eider-down comforters, sleeping bags, and padded jackets.

SYMBIOSIS—ANIMALS LIVING IN MIXED HOUSEHOLDS

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SYMBIOSIS, a term from the Greek, is what the biologist uses for the living together of two dissimilar organisms. In a broad sense it includes such diverse relations as the lice living on man and rats in his house, the union of an alga and a fungus to form a lichen, and the cross-pollination of flowers by hummingbirds.

The story of the burrowing owls of our Western plains living in amity with prairie dogs and rattlesnakes as one happy family comes to mind as an example. But "foolish nonsense" is how the noted biographer of North American birds, A. C. Bent, characterizes such stories. He then goes on to quote evidence as to what actually happens, and one can see how the story originated. The prairie dogs, which are really plump, dumpy, ground squirrels and not dogs at all, dig their burrows close to each other on the prairie in colonies which have come to be called prairie-dog towns, or dog towns or simply "towns." Burrowing owls also take up their residence in these towns, probably because they find burrows ready made and do not have to dig their own as they are quite able to do.

MODERATELY PREDATORY The owls may make an occasional meal of a young prairie dog, and a prairie dog may perhaps dine occasionally on owl eggs, but on the whole owls and dogs get along on terms of easy familiarity.

Sometimes when alarmed, both may scuttle into the same burrow for safety, but each has its own burrow. With the rattlesnake it is different. The rattlesnake may live in burrows in the dog town, but when it is hungry it eats owl or dog as occasion offers. While the picture of a happy family of owl, dog, and snake is a myth, the symbiosis of owl and dog, at least in the same colony, is striking.

In Africa there is a tiny falcon only about eight inches long which is called a pygmy falcon because of its small size. When Dr. Friedmann was studying the social weavers in South Africa, birds which nest in large colonies under a common roof they make in a savanna tree, he found these falcons occupying nest chambers in thriving weaver colonies. There was no friction between the weaverbirds and the falcons, and they were sometimes seen to sit side by side. When Friedmann collected three of these falcons he found bird remains in their stomachs but they were not remains of the social weavers. Apparently the falcons were feeding largely on small birds, but they did not molest the weaverbirds which had made the nests the falcons were using.

PARROT-DUCK-OPOSSUM MÉNAGE We occasionally find a mallard nesting in a tree, on an old crow or hawk nest, and there are ducks like the wood duck and the golden-eye, which usually nest in holes in trees, but a South American duck called the tree teal habitually nests in a parrot's nest. The parrots, called monk parakeets, make their nests in compact colonies in the branches of trees, so close together that they form a single mass. The tree teal's usual manner of nesting is to lay its eggs in one of the chambers in this apartment-house colony. At first the eggs are laid on the rough twig floor of the nest, but as the eggs increase in number a lining of down, plucked from the breast of the bird, is added until it may even extend out the entrance of the nest. Apparently parrot and duck both get along amicably in their pendant treetop cradles. An opossum sometimes also finds these parrot nests to its liking, though one wonders if it may not have a meal of young parrot or duck in mind. But be that as it may, in different chambers of a single communal nest of these parrots, parrots, a duck, and an opossum have been found.

On islets off the New Zealand coast lives a rather large-sized lizard called *Sphenodon*. It's rather well known by name, at least, for it is one of those relics of a formerly more widespread group which are called living fossils. It is also noted for its remarkable development of a pineal eye, the remnant of an important sense organ in ancestral forms, and formerly an organ some philosophers supposed to be the seat of the soul. But here we are interested in the fact that petrels swarm to these same islands to dig their burrows and lay their eggs in them, and it is in these same burrows that *Sphenodon* spends its daylight hours. Apparently the insect-eating *Sphenodon* and the oceanic-feeding petrels share the burrows amicably, adding still another example of a rather long list of dissimilar organisms whose lives are associated.

BIRD APARTMENT HOUSES

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EVERY NOW AND THEN in our press appear blasts against crowded living conditions in our cities, and the tenements where people are crowded together. Often there is the implication that this type of thing is unnatural and abnormal. And yet when we look about us in the bird world we see that gregariousness is a common trait. We have only to remember the great flocks of starlings and blackbirds in the autumn, or the massed flights of water fowl. Not only in traveling and in feeding, but also at nesting time birds may gather together, and some birds nest in such close association that the term "apartment houses" or "tenement" is really applicable.

The martins' house on our lawn with perhaps dozens of closely spaced rooms (some houses have as many as two hundred rooms) is a case in point. The neat martin house, made of boards, is a man-made thing, but even before the white man came to this continent, and before the Choctaw Indians hung up groups of hollow gourds for the martin colonies to use, the martins nested in colonies. Even in recent years certain colonies we might consider unprogressive have been reported as using such diverse nesting situations as among the boulders of a lake shore in Minnesota, and the closely spaced woodpecker holes which riddled a dead pine in Florida. And probably it was always thus. The martins liked company at nesting.

CLIFF DWELLERS, TOO Perhaps it would not be proper to consider a colony of bank swallows, each with a separate burrow in the same small cut bank and roofed with the same few square yards of turf of mother earth, as a real apartment house of cliff dwellers. But the term has been used in connection with a West Indian woodpecker, where a dozen pairs were nesting in a single dead tree, and "the trunk was a veritable apartment house." A similar situation exists with the naked-faced barbet of West Africa. This bird too makes a hole in a dead tree for its nest, like a woodpecker, and colonies of thirty to fifty birds may be found nesting in a single dead tree, while other dead trees nearby, apparently equally suitable, are untenanted. Colonies of hundreds of nests of cliff swallows, the nests touching and overlapping, may be under the eaves of a single barn, or as they used to be and some still are, on the sheltered side of a cliff. But as these birds had nothing to do with the making of the roof, perhaps these too do not deserve to be rated as apartment houses.

In southern South America there is a monk parakeet that makes a real tenement. It nests colonially in treetops, and the nests of sticks are placed so close together that they merge and form a single mass, up to nine feet across, in which each parakeet has its own nest. Similar to this is the palm chat. This West Indian bird is small and thrush-sized, dull in color, brownish with a streaked breast, nothing remarkable to look at, but it carries amazingly large sticks, little thinner than a lead pencil and as much as two feet and more long up to the top of a palm tree, and there it makes its bulky community nest.

BUILD NESTS CO-OPERATIVELY These stick nests, which may be four feet and more across, are conspicuous and regular features of the landscape in Hispaniola. The colony consists of four to eight pairs of birds, and each has its own apartment in the bulky structure, and its own passageway to the outside. But in the parts of the community nests that hold the individual nests together and cover them there are roughly defined passages running through the interlacing twigs of the top of the nest so that the birds can creep about under cover. Apparently some of the work is carried on in common, for as many as half a dozen birds may be working close together, pulling and twisting twigs more firmly into place.

The social weaver is the most advanced apartment builder. It, like the palm chat, has little of distinction in its appearance, being mostly dull brownish with a black face. But in its home country, the savannas of Rhodesia in southeastern Africa, its huge community nests in the savanna trees may be seen from afar. The largest Friedmann saw when he was studying the bird there was about 25 feet by 15 feet, by 5 feet high, and contained about 95 nests. And this might have been still bigger, for part of it had broken the branch on which it rested and fallen to the ground. Sir Andrew Smith, the early ornithologist of South Africa, has written that when these birds start a colony they first of all make a roof of coarse grass. The group to which the social weaver belongs gets its name from the remarkable ability some of them have of weaving their nesting materials. But the social weaver neither plaits nor weaves its roof. It puts the roof together in the form of a well-made hayrick with a fairly definite thatching arrangement so that the water runs off. This is a community effort. Under this roof each individual pair makes its own separate nest. These apartment houses are used year after year, but last year's chambers are not used, new ones being made under the roof each year, and so it grows bigger and bigger until the weight of the mass may break the branches and cause a part or the whole to fall to the ground.

BIRD HELPERS AT NESTING TIME

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IN MANY a well-run American home the children have definite responsibilities, the older children may help look after the younger, and even grown-up relatives may stay as part of the family group. As in so many cases there may be found parallels to this in the bird world.

The ani, the curious tropical American cuckoo that makes communal nests, is gregarious and the young of the first brood become part of the parent flock. Two more broods may be raised during one season in Cuba, and the young of the earlier brood may feed their younger brothers and sisters of the later brood. The same has been recorded for many other species in the wild: in eastern bluebirds, mountain bluebirds, wheatears, long-tailed titmice, barn swallows, coots, rails, and gallinules young have been recorded as feeding still younger birds. In captivity this habit has been seen a number of times. Young birds hardly able to feed themselves may help feed still younger individuals of the same or other species, and a nestling crowned hornbill has been seen to offer food to its nestmates. This tendency to feed nestmates evidently appears very early in the life of the bird, as Dr. C. O. Whitman, who worked intensively with pigeons at the University of Chicago, recorded a hybrid dove only twelve days old that fed its nestmate.

FIVE JAYS AT A NEST It was rather generally known that occasionally more than the two parent birds attended a nest, but until 1935, when Alexander

Skutch, the authority on the biology of Central American birds, published his paper "Helpers at the Nest," few of us realized how widespread this was. Since most birds of a species are difficult to identify individually, one must actually see the extra, unmated helpers at the nest along with the parents to be sure they are there. In the brown jays of Central America that Skutch studied closely the colors of the soft parts, bill, feet, and eye rings were variable and he was able to recognize many individual birds. At five nests he watched he found at least one helper at each nest, and at one there were five helpers, all bringing food. Sometimes, if between an incoming, food-laden bird and the young, they would take the food and pass it on to the nestling. At one nest the unmated helper was more zealous in guarding the nest than were the rightful parents. Sometimes, perhaps, these helpers were unmated young of the parents' previous year's brood, but this could hardly have been the case where there were five helpers, for the brown jay ordinarily raises no more than three young a year. A black-eared bush tit of Central America seems to have a great preponderance of males and at one nest in addition to the parents there were three other males bringing food to the young.

MATERNAL PENGUINS Perhaps the most striking example among birds is the emperor penguin. These birds breed in the dark and cold of the antarctic winter, on the edge of the ice shelf. The single egg is carried on the feet of the brooding bird; indeed one wonders what other adaptation for holding the egg would be possible in this land of ice, snow, and water. Only a few of the adults in each colony lay eggs any year, perhaps one in five, or one in twelve. But all the adults in the colony have the urge to incubate and brood. Thus many old birds, rather than merely the two parents, may take turns caring for each egg or chick, leaving the rest ample time to feed. So strong is the urge to brood that struggles may take place over a chick and it may be very roughly handled. Indeed the chicks may so resent this that they may creep away into ice crevices and freeze to death. Another strange turn this behavior may take is that frozen eggs, dead chicks, and even bumps of ice of suitable size are carried on the feet and covered with the birds' feathers by their "would-be fathers and mothers."

A NAME FOR A BOAT



A REQUEST for the name of a sea bird, a name to be used for a boat, came to me at my desk in the museum one day. My memory was quickly exhausted with sea gull, sea swallow, and albatross. But I keep within reach the handy guide, *Birds of the Ocean*, by W. B. Alexander. In the index I found twenty pages of names, two columns to a page. They started with *aalge*, *Uria*, and went on down through the alphabet to *yelkouan*, *Puffinus*, and to *zimmermanni*, *Sterna*.

EUPHONY NEEDED A name should be short, pleasant-sounding, and easy to remember and to say, so obviously such words as *Macronectes*, *Brachyramphus*, *Aptenodytes*, and *Coprotheres* are ruled out among the scientific names. But further, when choosing a name for a boat from among those of water birds, one should consider the kind of a boat. There should be some appropriateness; some points of resemblance between the boat and the bird, or between the boat owner and the bird. Albatross seems right for a seagoing sailing ship, sailing to southern oceans; tern (or sea swallow) appropriate for light, dainty coastal sailing craft; puffin or auk or murre for power craft, for these birds spend most of their time stolidly on the water and when they fly have a direct buzzing flight. Loon and dabchick would do well for fresh-water boats. But one objection to both them and the various auks for a name is that these birds spend much time swimming underwater. They might better give their names to submarines. The big,

stocky sea ducks, called scoters and eiders might suit some stout craft that ply to arctic waters.

SCIENTIFIC NAMES AVAILABLE I reviewed the host of other names. Scientific names need not be ignored either. What is nicer than *Gygis*, the name of the white, fairy, or love tern of the South Seas for a small summer sail boat? Then going farther afield into austral waters for far traveling craft there's *Diomedea*, the name of the albatross, and *Daption*, the medium-sized petrel that also is called pintado for the same reason a white-splashed horse is called a pinto, and *Prion*, the tiny whalebirds of the antarctic whose blue-gray back is near the ideal ocean-camouflage color. *Larus*, a good honest name without frills, belonging to the gulls that haunt our harbors, coasts, and lakes, would do for a plain, everyday sort of boat. Kittiwake is another gull that spends more time at sea. Gannets are boldly black and white, strong-flying birds of the North Atlantic, and one could use that, or its scientific equivalent, *Moris*, for a boat.

Penguin and pelican I'm doubtful about; I can't imagine a boat for either. Skua or jaeger would, of course, be a lovely symbol for a pirate vessel, as would frigate bird; both are birds that practice the stand-and-deliver method of getting food from weaker fisherfolk. The petrels called shearwaters are among the hardiest seagoing birds, but the name has little association for most people beyond wondering if they feed around breakwaters. Petrel itself isn't a bad name, though one might think of the storm petrels, which are also called Mother Carey's chickens, and have been considered the souls of drowned sailors, though their name perhaps refers to Peter, and his attempt to walk on the water, as these birds are continually trying to do.

Phalaropes are snipes of sorts that have taken up a periodic seagoing habit, and their name might often be appropriate. Even their habit of spinning quickly about as they sit on the water might still agree. A Chicago man named his Chris-Craft *Sandpiper*, after, as he said, the bird that goes hopping along the beach before the waves.

Sula is a good sort of a word, and the name of birds that are strong, swift fliers of the tropics. But in English they're usually called booby, which is an English word meaning simpleton (which name the birds got from stupidly perching on ships). *Alle* for the little auk or dovekie would do for a tiny

boat in northern waters, and I knew of one boat called the *Alca*, after the razor-billed auk, while *Cepphus*, the name of the black guillemots, is equally good, as is both *Lunda* and its equivalent puffin.

Some names have a stark simplicity that would attract few, like shag, used for the cormorant, and muttonbird for a petrel. The cahow people might shy from because for many years we were not sure whether this West Indian petrel was extinct or not.

Myself, there are two names I rather like and I've been saving for the last: for a small sailboat I'd say the *Wideawake*, as the sooty tern is called in its tropical home, and the other, for a larger seagoing boat, is the *Mollymawk*, a sailor's name for the albatross.

WEAVERS AND TAILORS IN THE BIRD WORLD

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ONE CAN IMAGINE the consternation in trade-union circles when it becomes known that there are, among birds, those who weave and those who sew. Their products are entirely for home consumption and there are no minimum wage, no maximum hours, or any fair-trade or quality agreements. None of the Audubon societies have even touched on the matter.

WEAVING The sewing and the weaving is done entirely in the construction of nests. To take up the weavers first, we can point to the Baltimore oriole, which makes a sac-shaped, pendant nest, often hung from the trailing tips of elm branches. The walls of this sac are formed of fibers pushed and pulled back and forth with the birds' bills in a seeming haphazard way so that a roughly woven or stitched fabric results. But the finest weavers belong to that group of birds known as weaverbirds. One might expect that to be an expert weaver a bird would have to have a slender bill. But no, their bills are short, stout, clumsy-looking, and sparrowlike. And yet these are the birds that weave elaborate pendant nests of fibers and straws. The finest are in shape like an inverted retort, with the nest proper in an oval chamber, fastened to a branch by a special strand of fibers, and with a tube or funnel for an entrance. The walls of these fine weaverbirds' nests are amazingly

strongly and neatly woven. In captivity one of the weaverbirds, the red-billed weaver, was studied at its nest building and it was found that the strong, intricate, and beautiful weaving of this species actually included knots of several sorts.

TAILORING The tailoring is done by birds of quite another group. They are Old World warblers of several sorts, some in southern Asia and some in Africa. The tailoring consists of sewing the edges of leaves together to form a place for their tiny nests. The Indian tailorbird is perhaps the best known. When these tiny olive-green and gray birds set about nest building the female punctures the margins of the leaves with her bill. Then she brings cobwebs and pushes them through the punctures in the edges of the leaves, and winds them around, and draws the edges of the leaves together. Strands of cotton are used too for this. Sometimes a single leaf is used; its two edges being drawn together to form a funnel. Sometimes a number of leaves are joined. Sometimes it is claimed knots are used, but this seems not to be the case. What are mistaken for knots seem made in this way: The cotton used is soft and frays easily, so that the part of it forced through a tiny aperture issues as a fluffy knob, which looks like a knot. "The bird makes no knots; she merely forces a portion of the cotton strand through a puncture," and the edges of the puncture catch and hold it, according to Casey Wood, who studied the birds in India. The lining of the nest is of soft material and this the bird anchors by making a puncture in the leaf, grasping a strand of this material, and pulling it out; the cotton outside then expands into a minute button which helps hold the nest and contents in place as though riveted. One nest is recorded as having been so riveted in seventy-five places.

The camouflage of the tailorbirds' nests is very good; it is usually built in thick foliage, the leaves are little deranged, the punctures do not cause the leaf to die; and the leaves being the same as the others, there is little for the eye to pick up as indicating a bird's nest.

SOCIAL PARASITES AMONG BIRDS

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THE MOTHER who would leave her infant on a stranger's doorstep, to be brought up an orphan, not even knowing its own parents, is a despicable character in human society. But when we leave the man-made society we must leave man-made rules of behavior and man-made prejudices behind. Morals are human. The rest of the animal world is not immoral, it is amoral. It cannot afford criteria beyond survival and reproduction. So while we call certain birds "social parasites," we attach no stigma to them. They represent several groups: the cowbirds, the weavers, the cuckoos, the honey-guides, and the ducks.

Carelessness in egg laying is common even in birds that ordinarily lay their eggs in their own nest and care for them themselves, as for instance the robins' eggs that you may find on your lawn (which of course are wasted; addling and rotting). Perhaps the fate of the eggs of pheasants and ruffed grouse which are found in the same nest may be more happy. Ducks usually make their own nests, but many species occasionally lay eggs in the nest of another species, and one South American duck no longer makes any nest of its own, but is a social parasite, not only on other kinds of ducks, but also on coots and some other birds.

The small, well-marked family of honey-guides of Africa, notable in other ways, also is remarkable for being social parasites. Their favorite host

species, chosen to look after the eggs and young, are their close relatives, the barbets (which themselves are most closely related to our woodpeckers).

The nesting of certain African weaverbirds was long a puzzle to ornithologists until it was found they too were social parasites, on other weaverbirds.

VARIED NESTING HABITS The cowbirds, of several species in North and South America, belong to a family notable for the variation in its nesting habits. Their nests vary from the elaborate purse-shaped structures of the oropendola and orioles to the dome-shaped nest on the ground of the meadow lark, the simple cup of the bobolink and redwing; the cowbird makes none. The cowbird lays its eggs in the nests of a wide variety of other species to be cared for by them. Here those who discuss the relative importance of heredity versus environment can profit by considering these social parasites. The young cowbird, hatched and brought up by, say, a yellow warbler, remains a cowbird. As soon as it no longer needs its foster parents' care it flocks with other cowbirds, with all their mannerisms and characteristics, and next season it mates with another cowbird. There is nothing left of its early environment.

The cuckoos of the New World and some of those of the Old make their own nests in normal avian fashion. But a number of Old World species are social parasites, and their behavior has long been a subject of study and discussion. Specializations indicate that here perhaps we have the highest stages of social parasitism. Whereas the cowbird may grow up with nestmates that are the young of the foster parent, unless perchance it crowds them out or starves them if it is larger, the young cuckoo gets the rightful occupants of the nest on its back and throws them out of the nest to perish.

EGGS LOOK ALIKE Another refinement in social parasitism by the European cuckoo is that apparently certain individuals, and apparently certain strains, lay their eggs only in the nests of certain host species. And these cuckoos' eggs resemble those of the particular species in whose nest the cuckoos' eggs are laid. For example, if certain cuckoos lay their eggs only in the nests of meadow pipits these cuckoos' eggs would resemble those of meadow pipits, while another group of cuckoos specializing in hedge-

sparrows would have eggs resembling those of hedge-sparrows. Another oriental cuckoo has a color adaptation in the young. In southern Asia these cuckoos parasitize crows, and the nestling cuckoos have black feathers like the young crows; in the Australian area where the same species of cuckoo occurs it parasitizes grayish-brown honey eaters and the young are brown, more like the rightful nestlings. Both these resemblances apparently reduce the chances of the cuckoos' offspring being rejected by the foster parents.

FISH EATS BIRD!

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IT HAS BECOME commonplace to hear about birds eating fish. The government gets out reports on the relation of fish-eating birds to fish abundance. The cries of commercial fisheries have caused inquiries to be instituted into the food of cormorants that were supposed to be eating the fish before they grew up enough for us to eat. The scarcity of salmon in some of our Northeastern streams has caused the allocation of biologists to study the predation of kingfisher and merganser on salmon fry and fingerlings.

But fish get some of their own back by eating birds. It's not as spot news as the "man bites dog" angle, but it's certainly less widely known.

To one who has fished for large-mouth black bass among the cypress trees and the bonnets of water hyacinth, and seen the bass strike savagely at surface lures as soon as they hit the surface, it comes as no surprise to find they strike at, and catch, such birds as Maryland yellow-throats that flutter across close to the surface of the water.

Young ducks, too, are good game to the large-mouth, and probably many a young duck finds its way into the maw of a bass. On a pond where bass had taken many young ducks I heard a story of a fisherman who made a floating model of a mother duck, powered it with a propeller, and attached to it by

lines of various lengths several floating models of downy ducklings. In each duckling was concealed a hook. The whole flotilla was set afloat, and drifted across the pond. Mother steamed ahead, with young following. Soon the bass, used to a duck diet, began to grab the ducklings. When the model was retrieved several large bass were taken.

In Northern waters, where Northern pike, or jackfish, as they're called in the North, abound in duck-nesting waters, pike are accused of eating enough ducklings to affect the survival of the broods. Many a marshland traveler has reported young ducks and young grebes diving, to be seen no more. He's blamed the pike. Sometimes perhaps the young bird has simply come up unobserved. But enough pike's stomachs have proved to have young ducks in them to demonstrate pike do eat ducklings. Strangely, in some areas, pike eat many ducklings; in others they do not eat them. But it's not alone young birds or small birds that get eaten by fish.

A twenty-four-inch bass is recorded as having been caught while it still had the legs of a full-grown coot projecting from its mouth. From beak to tip of its outstretched legs the coot measured seventeen inches and it weighed one and one quarter pounds.

Angler fish weighing between forty and fifty pounds have been found to have eaten birds. One had the band from a Manx shearwater in its stomach, and another had an adult American merganser. In tropical and subtropical seas the examination of birds seemed to indicate they had been attacked by some fish and seized by the feet, but were able to escape, and a white-winged black tern off Corsica has been seen to disappear under the water, presumably dragged under by a fish.

CROWS ARE SMARTER THAN "WISE" OWLS

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THE OWL has always been considered the symbol of wisdom. The old saying has it that "fine feathers don't make fine birds," but I'm afraid that the owl has taken in people with its appearance. The owl's reputation for wisdom seems to be based on a staid, impressive appearance combined with an inarticulate disposition. Though owls do at times make a great deal of noise, hooting, shrieking, and whistling, much of the time the owl sits quietly looking wise and saying nothing. But owls don't seem to have much behind the front they put up. People who have studied them find the young are very slow to learn to feed themselves, and one saw-whet owl that was kept captive refused to eat liver put into its cage, apparently not recognizing the meat as food. But when the liver was stuffed into an empty mouse skin the owl at once ate it. One might conclude that the owl was the original "stuffed shirt."

The crows and their relative, the jays, are the birds that are really intelligent. They are active and usually have little trouble getting enough to eat. They have an abounding curiosity that leads them to spend their time investigating things and getting new experiences. And they seem to profit by these experiences, too.

The following is how three ravens co-operated in getting a bone from a dog, as written by B. J. Bretherton:

"He was espied by a raven who flew down and tried to scare the dog by loud cawing, in which he was shortly afterwards assisted by another, both birds sidling up to the dogs head until they were barely out of his reach. Just at this time a third raven appeared on the scene and surveyed the situation from an adjacent fence, but soon flew down behind the dog and advanced until within reach of his tail, which he seized so roughly that the dog turned for an instant to snap at him, and at the same moment the bone was snatched away by one of the ravens at his head."

CROWS LEARN FROM OTHERS Crows have been recorded as profiting by the experience of one of their numbers. In Washington, when almonds were ripening in the almond orchards and crows were swarming there threatening to destroy the nut crop, an estimated 30,000 crows were involved and the destruction of an \$800 crop was complete in two days. Various methods of control were tried unsuccessfully. Finally some almonds were slit open, poisoned, and scattered about in the orchards. Very few crows were actually poisoned, not exceeding 1 per cent of the flock. The first reaction of the crows when one of their number was poisoned was one of extreme panic. There was tumultuous clamoring and confusion. Then the flock abandoned the attempts to feed on almonds and left the area completely. Here we have a case of superior intelligence, birds profiting by the sight of a few of their numbers being poisoned fleeing the area and so escaping being poisoned themselves.

TAME WILD BIRDS

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WE THINK of wild birds as being shy creatures by nature. For those of us who have kept a feeding station for birds in the winter so as to have the pleasure of association with the chickadee, nuthatches, woodpeckers, and other visitors, one of the most attractive things is that the wild birds become tame. Through association with persons they gradually learn that human beings are not to be feared. The high point of many a bird lover's experience is when a chickadee becomes so tame that it will perch on his body and without fear will feed from his hand.

It seems to be true that birds in wilderness areas are wilder and more shy of men than those living about dwellings where they are protected. This is notably true of the robin. In villages they hop around on the ground unmindful of the near presence of humans. How different they are in the wilderness, where the robins fly away apparently in great fear, while the human intruder is still far distant.

It comes as a considerable surprise to find that here and there over the world there are instances of birds with so little fear of humans that they come and perch on them.

PERCHING ON PEOPLE In the Galápagos Islands, where the general fearlessness of birds is famous, one of these cases is recorded. David Lack, who was

studying the biology of the Galápagos Islands' birds, found when walking through the woods on Indefatigable Island that a flycatcher would sometimes try to settle on his head. Lack stood still and found the bird's object was to pull out some of his hair. The bird, having failed to detach any of the hair of his head, tried, apparently with no better success, to pull out hair from his eyebrows and then from his chest. This was at the height of the breeding season and apparently the bird was trying to get nesting material. This seemed to be a usual type of behavior there, and Lack correlated it with the general tameness of the birds on the islands.

There is a honey eater in Australia that includes in its pattern of behavior perching on people's heads and shoulders and attempting to pull out hair for use in its nest. A. H. Chisholm writes of going to certain places and taking companions with him for the sake of experiencing this, and the practice is so common with the species that Australians refer to this honey eater as "the hair dresser." In this case it is not tameness alone. The white-eared honey eater, which indulges in this practice, is no more tame most of the time than any of the other small local birds that live in that part of Australia. Only at nesting time does it attempt to light on persons. Chisholm correlates this hair-plucking trait with other habits of the honey eater: he speaks of its gathering hair from such animals as rock wallabies and gathering bristles from farmyard pigs and goats.

Our familiar phoebe has been recorded as perching on deer hunters in the fall and using them as a vantage point from which to conduct its hunting. This was in North Carolina, and the weather being warm, mosquitoes were notably in evidence. The bird showed no sign of fear or nervousness, but perched on the hunter's gun, on top of his head, and various parts of his body, and then flew out and picked up mosquitoes. As the hunter's face seemed to be attracting more mosquitoes the phoebe directed his attentions there. In picking mosquitoes off his face the sharp points of the bird's bill were noticeably felt at every capture and the irritation caused by a succession of these pricks caused the hunter to decide that he could take care of the mosquito situation better without the help of the phoebe. As H. H. Brimley, the hunter, put it "... my face was beginning to feel somewhat inflamed from the frequent pecks to which it had been subjected so I called it a day and told the phoebe to stop pestering me." This took place in a wild

part of North Carolina and Brimley suggested that the phoebe's abnormal lack of fear was caused by its having never seen a human being before.

BIRDS AS PILFERERS

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PILFERING, or petty theft, is one of the less desirable but very human attributes of our race. But it's also pretty widespread in the animal kingdom. Theft as the usual thing is practiced by only a few birds. But when it's a case of petty theft, happening now and then, it is common enough in the bird world. It's not restricted to any group of birds, but may crop up almost anywhere. There's no threat or fight about it usually. The bird, which gets its food by means of the acuity of its vision and the quick co-ordination of its movements with the recognition of its food, sees the food in another bird's possession and just goes up and takes it. Sometimes the food is taken from a larger and stronger bird, an achievement accomplished by audacity, agility, and quickness. A sparrow hawk, that inoffensive little rufous-red falcon that spends most of its time catching grasshoppers, was sitting on a telephone wire holding a small mammal it had caught, apparently about to devour it, when a loggerhead shrike sitting nearby flew straight to the hawk, seized its prey, and made off, leaving the hawk sitting there, apparently dumfounded by the audacity and success of the attack. A case in which the pilfering caused a mild fuss involved an English kingfisher and a dipper. The kingfisher lit above a pool where a dipper was feeding, obtaining food in the pool and bringing it ashore to eat it. When the dipper next came ashore the kingfisher flew down, there was a momentary scuffle, and the dipper departed, leaving its food to the kingfisher, who promptly ate it. Despite

this occurrence the dipper allowed itself to lose its prey again before it left, and the kingfisher presumably had to resume fishing for itself.

THEFT NOT RESENTED It is sometimes surprising that this pilfering, when it occurs over and over again, is not actively resented, particularly when the pilferer is a smaller bird. Some of the thrushes are especially docile when they're victimized. Sometimes when American robins are feeding on the ground, house sparrows hop along with them, and when the robin finds a worm the sparrow hops up quietly and boldly takes the worm from the robin with scarcely a protest from the victim. One robin is reported to have been robbed six times, of six worms, one right after the other by a small flock of sparrows while the robin continued to hunt for worms.

The starling, an aggressive Old World species introduced and very successful here, also victimizes the American robin. In one case a starling made four successful raids in five minutes, the robin not attempting to fight or defend its food, but simply moving off a little way and continuing to hunt for worms while the starling waited nearby.

This is not a new trait of the starling, for in its Old World home, in Britain, it has been seen to victimize blackbirds and song thrushes (relatives of our robin). This happened when a blackbird pulled up a worm, a starling flew to the spot, and the blackbird moved away, leaving the worm to the starling. This method of obtaining worms was sometimes used by all the starlings on a lawn where both species were feeding, much to the hindrance in the feeding of both blackbirds and song thrushes.

Gulls have been recorded as snatching fish from mergansers that had caught fish by underwater dives and brought them to the surface to eat. Gulls also follow pelicans, and just after the pelican has completed its plunge and before it can swallow the fish protruding from its bill, a gull may flutter in, alight on the water or even on the pelican's head, and seize the fish. The pelican does not attempt to do anything about it, but accepts the pilfering with stoic calm.

Grackles victimizing ibises seems perhaps the strangest of the whole series of reports. The ibis often attempts to elude the grackles but without success. About Lake Okeechobee, Florida, where ibis are common, they feed largely on crayfish, which they secure by probing the holes made by these

creatures. Grackles swarm there, and, on occasion, no sooner does an ibis seize a crayfish than one to four grackles try to secure it. The ibis may take flight and attempt to escape with its prey, but one of the grackles usually gets the crayfish away from it.

Possibly some of these birds are on their way to becoming habitual pilferers, in which such social parasitism is a fixed mode of life. With evolution, if this thieving benefits the species that snatch the food, it may become a usual habit. For habits, like structures, are subject to variations, to selection, and thus to change and elaboration.

HIBERNATION IN BIRDS

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NOT UNTIL 1948 did the scientific world have satisfactory evidence that any bird hibernated. True, it was an established fact that sometimes in cold weather some birds, notably swifts and hummingbirds, might become torpid for a short time, but this was not hibernation.

The early literature, of more than a century ago, contained many accounts, some claiming to be firsthand, of birds hibernating. Swallows in particular were reported as seen to submerge in ponds in the autumn. Numbers of them were said to have been found hanging to submerged willow branches apparently sleeping the winter away. When ponds were drained in winter, sometimes swallows were said to have been found buried in the mud, revived, and upon occasion kept alive indoors until the spring. Sometimes slime-covered swallows, evidently just out of hibernation, were reported found in the spring. Swallows were the most commonly recorded, but other species, too, were mentioned as hibernating, such as the cuckoo that shed its feathers and crept into a crevice to sleep away the winter.

Such accounts gradually disappeared from the literature. We can accept none of them. The old records of underwater and also the featherless hibernation of birds must be discarded. The occasional torpidity, in cold weather, of swallows, swifts, and hummingbirds is another matter, and appears to be of sporadic though not common occurrence.

FROGS MISTAKEN FOR BIRDS It is interesting to speculate as to how the old "firsthand" accounts originated. They had certain basis of fact. The first was that swallows were seen flying about in summer. They disappeared in winter. Aristotle claimed they hibernated, in a featherless condition, so there was nothing unusual in seeing them that way. Observation was less critical, and it is probable that frogs from the mud of ponds were mistaken for naked swallows, and perhaps bats, which do hibernate, taken from caves or hollow trees, were also mistaken for swallows.

AN AUTHENTIC RECORD In 1948, and again in 1949, Edmund C. Jaeger, of California, published accounts of a poor-will he found hibernating. This was the first acceptable evidence that such a thing occurs. In a little cavity in the wall of a canyon in the Chuckawalla Mountains of the Colorado desert in California, Jaeger found a poor-will in a state of profound torpidity in December, 1946. He could pick out the bird in his hand, examine it and put it back in the little cavity it occupied without eliciting more than a slight movement of its eyelids. On a later occasion handling it revived it somewhat.

The next winter Jaeger found a poor-will, perhaps the same bird, hibernating in the same niche. Over a period of eighty days, from November 26, 1947, to February 14, 1948, he visited it periodically, examined it, and took its temperature. The body temperature was low, 64°-68° F., compared with more than 100° F. of an active bird; with a medical stethoscope he could detect no heartbeat, and a cold metal mirror held directly in front of its nostrils collected no moisture from its breathing. The body processes were evidently very low. The bird was banded for identification, and in the third winter the same bird wearing the same band was found to have returned to hibernate again in the same rock niche. But on subsequent visits it was missing—perhaps having lived out its allotted span, perhaps the prey of some predator.

SNAKESKINS IN BIRDS' NESTS

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THERE ARE occasionally discovered behavior patterns of birds that are so unusual as to make one stop and wonder. They are unusual for birds generally, but in a species here and there they are the regular thing. Such is the placing of a shed snakeskin in their nests by some birds.

A bird like the English sparrow, or the road runner, which uses a variety of material coarse or fine, would be expected to use shed snakeskins occasionally, as it came across them. But there are a number of species that seem to use snakeskins regularly in their nests. It would seem that the birds deliberately sought out the skins for this purpose, as though they were as much a part of the nest as the mud in the bottom of a robin's nest or the fresh green grass heads ornamenting the entrance to some weaverbirds' nests.

SOME HABITS BAFFLING I have long since given up thinking that every aspect of a bird's life must serve a useful purpose. Indeed I have already pointed out some definite maladaptations. But usually every type of behavior has a logical origin. By considering its occurrence in various species and against the background of the bird's everyday life some correlations usually can be found.

The list of birds habitually using snakeskins in their nests is short, as follows:

1. Great-crested flycatcher—belonging to the New World flycatchers, breeding in Eastern North America and nesting in holes.
2. Arizona crested flycatcher—a relative of the great-crested variety, with similar habits.
3. Blue grosbeak—an American member of the sparrow family, making an open nest in bushes.
4. Black-crested titmouse—a member of the chickadee family, living in Western North America and nesting in holes.
5. Bank mynah—a starling, living in southern Asia and nesting in holes in banks.
6. Rifle bird—an Australian bird of paradise, making a cup-shaped nest in trees.
7. Madagascar bulbul—making a cup-shaped nest in trees.

LIKE A DECORATION Twenty or more other species of birds have been recorded as using snakeskins more or less commonly, or occasionally perhaps on the basis of availability or of chance. But with the above they're an essential part of the nest. In some of the species the snakeskins are arranged as a rim around the edge of the nest almost as a decoration; sometimes the snakeskins may make up most of the nest.

Now as to possible correlations. The species are not closely related. Except for the two flycatchers the other five represent five different families. The distribution over the world is wide, too: America, Asia, Madagascar, Australia. Various explanations for the behavior have been advanced. It has been suggested that it's correlated with hole nesting, but three of the seven do not nest in holes. The most common theory is that it's to frighten away possible predators by making them think there is a snake in the nest. However, this is not very likely, and, too, one wonders why the birds that use the snakeskins are not frightened themselves. Indeed, one writer, surely not seriously, has suggested that the fright in early life of crested flycatchers at finding a snakeskin in the nest accounts for the upstanding crest in this species!

"BURGLAR ALARM" THEORY Another suggestion is that the snakeskin, by the rustling noise it makes when touched, acts as an alarm bell or a burglar alarm to warn the rightful occupants of the nest when an intruder approaches. This also seems a rather weak explanation.

We are left, then, with the fact that this curious habit has been developed in a few birds, not closely related, that live in various parts of the world and that have very different habits. It is usual with them. A number of others occasionally have this habit.

My first clue as to the proper background against which to solve this habit came when, unpacking a bird collection made in Borneo by curator of anatomy D. Dwight Davis, I took out a bulbul's nest. In its outer edge were flat, weathered leaves that resembled snakeskins. Later, when we received a bird collection from Dr. D. S. Rabor of the Philippines there was a nest of another species of bulbul and this too had flat, dead, weathered leaves in it that looked like snakeskin. When I was in Madagascar, in 1929-31, I had found three nests of the Madagascar bulbul with a snakeskin used in each. Here was a clue. I decided to investigate the nests of the other species of bulbuls of southern Asia and Africa where the family is represented by many species. By considering the snakeskin-using species against the background of the nesting of the other species some correlation might appear.

BOOKWORK This became a library problem at once. I had to look up the earlier reviews of the problem in the ornithological journals, *The Auk* and the *Ornithologische Monatsberichte*, then in Strong's *Bibliography of Birds*, to make sure that no important papers were missing from my own subject file. Stuart Baker's *Fauna of British India, Birds* had a large part of one volume devoted to bulbuls, and gave excellent summaries of the nidification of each species occurring there. Bannerman's *Birds of Tropical West Africa* covered the western part of that continent, and Jackson's and Sclater's *Birds of Kenya Colony* did the same for the eastern part. For collateral material I looked in Mathews' *Birds of Australia*, Volume 12, Forbush's *Birds of Massachusetts*, and Mrs. F. M. Bailey's *Birds of New Mexico*, and a dozen minor publications.

But it was worth it.

Perhaps my earlier thinking was dominated by the thought that the shed snakeskins had been parts of animals toward which many birds show an antipathy. But it's extremely probable a bird does not recognize the snakeskin as such. Rather to it the shed snakeskin is a strip of thin, flexible material. Obviously it would be used, by chance, by many bird species, such as the house wren, which, in addition to using such natural materials as twigs, grass, and hair, has been recorded as using lead pencils, paper, nails, safety pins, and snakeskins in its nest.

As to the regular users of snakeskin, the snakeskin-using Madagascar bulbul did fit into a pattern. Bulbuls in general make characteristic simple cup nests. Some species use almost any available material. But quite a few species had specific choices of materials: one species' nest had tendrils of vines in its base; another a lining of grass heads of a certain color; another pine needles; another red dead leaves; and the Madagascar bulbul snakeskins.

A SOLUTION There seems to be a tendency for many species to make distinctive nests. They often accomplish this by a choice of material used by few or no other species. What more natural than that one species, being in a country where snakes are common, should hit on shed snakeskins!

To show that the choice of snakeskin as nesting material is an expression of a tendency for each species of bird to make a different kind of nest may not be much of an answer. But it is to an extent. No longer do we say, "Why are certain birds' nests characterized by snakeskins?" Rather we have the broader, more general question, "Why does each kind of bird tend to build a nest different from that of every other kind?" Thus, little by little, we clear away small, vexing questions and resolve them into larger, more general questions. For answers to these we sometimes plan extended work involving field studies, studies of specimens, and books. And sometimes, as we examine a specimen, read a paper, or unpack a shipment, an answer, or at least a clue, springs to our mind.

CO-OPERATION BY BIRDS

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THE IMPORTANCE of co-operation, contrasted with competition, has assumed increased importance in discussions of evolution, as it has in discussions of human social progress. Co-operation in nature is of various kinds; from the manner in which a forest shelters the squirrel to the manner in which two or more individuals of one species work together for a common object. The working together of two birds to rear a family is so well known an affair that one forgets that it is an example of co-operation, not only in building the nest and brooding and feeding the young, but also in defending the nest and the young.

Sometimes more than one species will join in ousting an enemy. For example, when a cat caught a young robin, recently out of the nest, the parents, in their frantic effort to make the cat release the bird, attracted the attention of another robin and a pair of cardinals nesting nearby in a honeysuckle. All five birds dived at the cat, screaming and pecking it so vigorously that it released the young robin and fled.

EAGLES JOIN EFFORTS More spectacular are some of the co-operative activities of birds in food getting. Bald eagles sometimes feed on ducks. Frequently two eagles may combine their efforts. The two birds may work together to force a black duck from the air onto the water, and when they are trying to catch a diving duck, they much more quickly exhaust their prey by

swooping at it in turn. Bald eagles sometimes take water birds too large for them to carry, and then they must flap along dragging their prey on the surface of the water to the nearest shore. On one occasion an eagle dragging a large cormorant ashore was joined by two other birds, and all three took turns in dragging it. When they got it ashore, all three shared it.

Several fish-eating birds co-operate in capturing their prey. "The merganser is primarily a fishing duck ... very skillful and a voracious feeder. It pursues underwater and catches successfully the swiftest fish. Often a party of sheldrakes may be seen fishing together, driving the panic-stricken fish into the shallows or into some small pool where they may be more easily caught," according to A. C. Bent.

When a school of fish approached a flock of white pelicans, the birds suddenly assumed a circular position, surrounding the school. All the pelicans moved slowly but cautiously toward the center of the circle, their heads near the surface of the water or partly submerged and their necks slightly extended. The birds moved in perfect unison, making the circle progressively smaller, ready to engulf their helpless victims at the first opportunity. When all the pelicans were close to the fish, the birds made rapid jabs at the fish and apparently consumed a large number of them. It appeared that every bird got from one to several fish.

13,000 BAND TOGETHER Avocets and, to a lesser extent, the black-necked stilts also band together for co-operative drives on small fry and aquatic insects. Such drives are made in water of wading depth. Instead of forming circles the birds present compact spearhead and wedge formations and sweep the bottom muck with the characteristic back-and-forth side movements of their long bills. As many as 13,000 avocets have been observed taking part in such co-operative feeding projects.

Another striking example is furnished by black vultures observed by E. A. McIlhenny. A three-quarters-grown skunk was wandering across a field. A vulture alighted near the skunk which immediately stopped and raised its tail. Other nearby vultures joined the one nearby the skunk, and when six or eight of them had gathered one suddenly attacked it. The skunk immediately discharged its defensive scent, but without effect, for the vultures attacked in a mass and other vultures circling above joined in until

there were probably twenty-five or more around the skunk. With much flapping and croaking, the vultures pulled it about until it was dead, and then devoured it.

On another occasion a black vulture came from high in the air to alight near two full-grown opossums following a narrow cattle trail. The first vulture was almost at once joined by many others until there were probably between seventy-five and one hundred black vultures following the opossums. Suddenly three or four of the vultures attacked and the others joined in. Quickly both opossums were covered with a swarm of hissing, flapping birds, and within fifteen minutes there was nothing left of them but the larger bones and the hides, and these were stripped of every vestige of flesh.

WATCHDOGS AT THE NEST

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A SAVAGE WATCHDOG outside his master's house helps to protect it. If an intruder comes, the watchdog, if it's the right kind, simply bites him without preliminaries. There's a parallel to this in the bird world. Some birds often have their nests close to wasps' or bees' nests, or in trees inhabited by biting ants. The birds and the ants, wasps, or bees get along without disturbing each other. But when intruders come along the insects swarm out, biting or stinging and driving the intruder away. The insects are protecting their own homes, but one of the results, the protecting of the birds' homes, is just as satisfactory to the birds as if they did it on purpose. This building of birds' nests close to wasps' nests is a common practice with certain sunbirds and weaverbirds, especially in Africa. It occurs too often to be chance. The question naturally arises as to how much the birds understand of it all—do they actually seek out the association? That's difficult to say, but the facts of the association are still there.

Though some of these associations are evidently fairly common and chosen deliberately by the birds—and one can easily visualize how the protection works—field observations as to the natural enemies against which they are effective, and how effective they are, are largely lacking. Usually the records are something like those of Van Rossem for the Giraud's flycatcher in El Salvador, in which he points out that this bird usually nests in certain mimosa trees armed with numerous heavy, curved thorns. These thorns are

hollowed out and inhabited by swarms of small but extremely hostile antlike insects, so that the nest is well protected. However, the effectiveness of ant and bee protection against human predation can be seen in the following.

Take the case of Mr. M. E. W. North, who arranged a rope to climb to a fish eagle's nest in East Africa. He had gotten about fifty feet up and was considering going out on the big limb on which the nest was, when he noticed a wild bee on his sleeve. Realizing that he was disturbing a wild-bee hive, and knowing that the sting of these vicious bees can be dangerous, fatalities having been reported, he came down his rope at express speed, crashing through projecting branches and brambles. Reaching the ground, he freed himself from the rope and fled to a safe distance, considering himself lucky to have received only three stings.

On another occasion, again in East Africa, Mrs. R. E. Moreau attempted to reach a hawk's nest to measure the eggs, but when she was up in the tree, savage, biting red ants drove her out.

BIRD GUIDES TO HONEY

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IN AFRICA there are birds which lead men to honey. They are called honey-guides and their family name, Indicatoridae, has the same idea incorporated into it. Though there are several species of these small, dull-colored birds, which are related on the one hand to woodpeckers and on the other to barbets, it is only one species, the common or black-throated honey-guide that is well known as a guide to honey.

The traveler in the country may find one of these birds chattering and flying ahead of him. The natives, who know this bird well and favorably will tell the traveler to follow; it will lead to a bee tree. The native, as he follows this guide, gives occasional whistles, as if to encourage the bird. The bird continues, flying from perch to perch, ahead, and chattering noisily. Sometimes it may return to see if the men are following; sometimes it remains chattering on its perch until the followers catch up. Finally the bird will go no farther. It flies about aimlessly and allows one to approach closely. This is the spot. In a hole in the tree trunk, or in the ground nearby the bees' nest is to be found.

When the beehive is opened, and the honey taken, the honey-guide will eat the comb that is left, and apparently it is for this that the complicated behavior of leading of man to the beehive is developed.

Wax of the honeycomb is a usual food of this species, judging by stomach examinations, and one wonders how they get it when man is not about to open the bee trees for them. The birds have no special adaptation for getting into the hives; indeed their only apparent adaptation for this habit is a thick skin, perhaps a protection against bee stings. Perhaps, as has been suggested, other animals, squirrels, monkeys, or honey badgers may unwittingly aid them by opening up bee trees for their own purposes and allow the honey-guides to snatch food for themselves.

An amusing side of the picture is that sometimes the honey-guides may lead the honey hunter to a beehive owned by a native.

There are also records of the honey-guide leading men to big game: leopard or lions. That this occurs is amply documented, but one wonders whether or not this was accidental; the honey-guide leading the way to honey perhaps by accident leads the way past the resting place of one of these big cats so that the man stumbles over the big game and perhaps gets the impression he was led to the animal.

OXPECKERS

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THE LIVES of oxpeckers are so linked to those of large, hoofed game or domestic cattle that in West Africa where game is scarce the birds depend on cattle, and their range is restricted accordingly. There the cattle are confined to the higher and more northern areas, free of tsetse flies, from Senegal to Northern Cameroon. Thus tsetse flies help to determine the limits of the oxpeckers' range.

Except for their nesting, which is in holes in trees, and their sleeping, most of their time is spent on the bodies of the larger herbivores. There they run about over the hides and legs of the beasts, like woodpeckers on a tree. They stay remarkably close to the animals, and even ride on them as they travel. The oxpeckers' food is largely ticks, which it gets from the hide of the animal by working over it with the side of its bill, shearing off the ticks with a scissorlike action of its mandibles. But when an animal has sores or cuts or scratches the oxpecker may peck into them, and eat flesh and blood of its host.

Correlated with this unusual and close relationship, a modification in the oxpeckers has taken place. There are only two species, both African, and they are dull-colored, modified starlings. The legs are stout, with curved, very sharp claws for clinging to the hides of animals, and the bill, very sharp at the tip, with the cutting edge of the mandible very sharp to aid in scissoring off ticks.

All the larger herbivores are attended by oxpeckers except the elephant and the hippo, but the favorite seems to be the rhino, and for this he's sometimes called the rhino bird as well as tickbird and oxpecker. The rhino gives the bird its food, and in return the bird provides a service of a value difficult to evaluate. It acts as a sentinel and may warn the rhino of the approach of hunters, for which habit it is execrated by sportsmen.

It would seem that such relationships could have developed only where the supply of big game was large. With the introduction of cattle and other domestic animals it was natural the oxpecker should turn its attention to them. Here the question arose as to the attentions of the oxpeckers being harmful or otherwise to the herds. Mr. R. E. Moreau, formerly of the East African Research Station at Amani, has investigated the problem. He finds that white men who own herds tend to consider the oxpecker a nuisance; Africans tend to consider it beneficial and some African cattle owners object to having the birds killed; the beasts themselves tolerate the birds.

There is the possibility on the one hand of oxpeckers spreading certain cattle diseases that are mechanically transmitted, and on the other hand they may help reduce disease by eating ticks, the vectors of certain diseases. Of course dipping the cattle takes care of ticks on them, and here we see another indirect effect of civilization on bird life. When cattle have been dipped the oxpeckers disappear from the herd. Perhaps it is because there is no longer food for them there; perhaps they get enough of the poison dip left on the beasts' hair to be lethal.

WINGS IN FEEDING

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THE OBVIOUS ADAPTATION of a bird's wings is for locomotion; to fly in the air. It is true that some few birds are flightless, and some like the penguins use their wings for underwater swimming, but this does not spoil the generalization.

Secondary uses, some with special adaptations, occur: the owl at bay spreads its wings wide, with the effect of increasing its apparent size and being more terrifying to a predator. The young bird, begging to be fed, flutters its wings in a characteristic way, and the female, in some of her mating behavior, may also flutter her wings like those of a young bird.

In courtship the wings may play an important part in display. In the Australian rifle bird they are held out, fully spread on each side of the bird like a velvet curtain against which the vivid iridescence of the throat patch stands out more vividly. The argus pheasant has the inner secondaries greatly elongated and ornamented in a fashion recalling the decoration of a peacock's tail and these he spreads to show in his courtship, while the ruffed grouse uses his wings to make instrumental music, his drumming.

Wings in geese and swans may be used in fighting, and tame birds may severely buffet humans who take too close an interest in their young. In the related screamers of South America the bend of the wing is equipped with long, very sharp spurs, which undoubtedly make formidable weapons in fighting.

In addition wings are used in at least three different ways in feeding. The red-tailed hawk may spread its wings as it sits on its prey, perhaps a behavior adapted to help the bird maintain its balance when dealing with struggling prey, perhaps to help smother the struggles of its prey.

The secretary bird of Africa is said to feed on snakes, poisonous and non-poisonous ones, and is said to use its huge wings as shields for its body in attacking them.

But the strangest use of wings in feeding is that practiced by a blackish African heron. In feeding in shallow water it takes a few rapid steps, apparently to bring it within reach of fish it has sighted, then spreads its wings, bringing them forward until they meet, and with the tips of the quills in the water. The head is in the canopy formed by the wings, and apparently it is here under this canopy that the fish on which it feeds are caught. The suggestion as to the correlation that presents itself is that the dark canopy thrown over the fish confuses them and makes them easier to catch.

INSTRUMENTAL MUSIC OF BIRDS



VOCAL MUSIC bulks large in our avian springtime chorus, but don't overlook the instrumental music that accompanies it. The drumming of the downy woodpecker on the dead limb of a maple near my bedroom window is as much a part of my spring as is the cheery cheerup of our robin. It's not that woodpeckers are voiceless that they drum. The flicker can be called in with his particularly rich repertoire to repudiate it vociferously. All day the downy woodpecker goes about pounding his head against tree trunks, with his bill chiseling out wood-boring insects to eat. What more natural when springtime comes and he wants to tell the world, and especially other woodpeckers about it, to select a dead limb with a nice tone in my maple tree and hammer out a rolling tattoo—his love song and his challenge.

A DRUMMER The gray-brown ruffed grouse of a wood lot we used to have in the Chicago area is a drummer I miss. "Thump-thump ..." he started slowly, and then quickened to a roll that filled the forest with hollow sound and you wondered whence it came, unless you happened to know, as I did, that an old log in the patch of gray birch was the old cock's favorite performing stand. There he came to roll out his invitation to the demure hen grouse. A drummer, I've called him, yet he has no drum. It's his wings, striking the air, that thump and build up into a roll, its volume testifying to his great breast

muscles as well as does the whirl of wings as he hurtles away through the air when I come too close.

The snipe of a nearby marsh makes music with feathers and wind, music that is more enthralling to me than the song of the yellowthroat or the vocal imitation of stake driving by the bittern. Circling high, then with a change of pace, his "winnowing" or "bleating" spring song comes drifting down. There is still room for argument, but probably it's air rushing past the outer tail feathers that makes the sound. One year a short-eared owl nested in the nearby meadow. Owls generally are vocalists, even if we don't rate very high their hoots or yelps, but the short-eared owl also has an instrumental performance. Sometimes, when giving his mating song on the wing, a series of "toots," he interrupted this by a dive in which he brought his wings together under his body, with a clapping sound. It's part of the performance, but not, as might be said, the owl applauding his own show.

Over our public school each evening in early summer a nighthawk booms. He has a voice, and he uses it, calling "beep" as he circles high. But the climax of his performance is instrumental, wind on feathers. He heads down, wings high, toward the flat gravel roof on which his mate is sitting. As he approaches the roof he moves his wings down; the air rushing past the quills gives a tearing boom as he comes out of the dive and mounts skyward again.

At dusk, at a damp corner of our old wood lot, in the spring, I listened for the woodcock's flight song, a twittering of wing music as he circles up, and sweet music, too, for a wild fowler's ears, is the whistling of the wings of a passing pair of black ducks on their way in the early darkness.

CONDITIONING IN BIRDS

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THE CLASSICAL EXPERIMENT in conditioning and reflexes is that of Pavlov. It consisted of sounding a bell each time food was given to a dog. Finally the salivary response resulted even when the bell was rung, without the food being given to the dog. The dog was *conditioned* to the bell. First it had responded to the food, then to the food and the bell, and finally to the bell alone, by a flow of saliva. The beauty of this experiment is in its simplicity, dealing as it does with a single reflex.

Though much behavior is more complex, experiments have been worked out to show how the environment, in a broad sense, can influence inherited behavior. An illuminating example of this is the one I made dealing with young loggerhead shrikes and the duration of their infantile behavior. Young shrikes, as with young passerine birds in general, while in the nest are fed directly by the parents, who place food in their mouths. One of the earliest behavior patterns these young birds perform is to stretch up with widely opened mouth, fluttering wings, and buzzing calls, in anticipation of being fed. This we call begging. Though typically infantile behavior, it may reappear in courtship, but this latter we will not consider here.

Ordinarily this infantile begging behavior is discontinued shortly after the young birds leave the nest and become able to feed themselves. Observations indicate that in a state of nature this change is probably hastened in part by the young birds themselves, who come to avoid having

food thrust down their gullets, and prefer to pick up the food for themselves, and in part by the waning interest of the parents in the young, which confers an advantage on the young who early become self-supporting.

CASE OF RETARDED DEVELOPMENT Certain observations made from time to time have indicated that though the age at which young birds changed from infantile begging for food to self-supporting independence was a fixed thing, started by instinct, certain external factors, notably the amount of care the young received, could affect the age at which this change occurred. Indeed there was a record of a young cedar waxwing raised by hand who never learned to feed itself.

When I secured a brood of four young loggerhead shrikes, or butcherbirds, the material was available to conduct a controlled experiment. The young birds were raised together by hand to the stage where they were ready to begin to pick up things, to feed themselves, and to begin to abandon their infantile behavior of begging for food. This was when they were twenty-one days old. They were then divided into two lots and housed separately. One couple had a supply of food kept in front of them, and hand feeding was gradually discontinued and stopped as soon as possible. At the age of twenty-eight days they fed themselves well, though they still begged freely when I approached. By the time they were thirty-nine days old they begged rarely, and after the age of forty-five days they were not seen to beg.

The other couple had no free food available at any time, and they were fed completely by hand, the food being placed in their mouths. At the age of twenty-eight days they had made no effort to feed themselves. By the time they were fifty-three days old they made efforts to feed themselves, trying to peck the food from the fingers instead of having it thrust into their mouths, and evidently would have changed quickly to independent self-feeding and abandoned their infantile begging behavior. But hand feeding was continued. At the age of seven and a half months, when the experiment was discontinued, though these birds were capable of feeding themselves, as was seen when food was accidentally dropped on the floor of their cage, they still begged for food from their human foster parent.

OBJECT LESSON FOR PARENTS These four birds used in this experiment were nestmates, and had similar heredity and early environment. The birds in the lot which received only enough care to ensure proper development became self-feeding, independent, and lost their infantile begging behavior when they were about a month and a half old. The other lot, which received an excessive amount of care in the latter part of infancy, and were hand fed without being allowed to develop the behavior that would have made them independent, retained the infantile behavior pattern of begging to be fed until the end of the experiment. They were then seven and a half months old, and their nestmates, under a different set of conditions, had lost their infantile behavior six months earlier.

With some birds it appears excessive care can be a conditioning factor. It can delay the loss of infantile behavior and the acquiring of the normal independence. Though instinctively the young shrikes tried to develop their independent behavior, when this was not possible they continued their dependent, conditioned behavior.

POISONOUS BIRDS

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POISON WE KNOW perhaps best in the plant world, whence comes, for example, strychnine. The deadly nightshade, a common weed, is another well-known poison plant. In the animal world we know poison best as something that is injected into the body by stings of bees, bites of spiders, the bites of insects, and even bites of shrews. In addition some animals having irritating, bad-tasting, or poisonous secretions which presumably protect the possessor from predators. This has received most attention in the insect world, the bad-tasting grasshoppers being examples. Toads have an acrid secretion from their skins which deters many would-be toad eaters, and pickerel frogs have somewhat the same thing.

The following three birds, which are recorded as having poisonous flesh, are, strangely enough, all members of groups ordinarily considered good table birds. Further, it seems the poisonous properties of their flesh are not constant, but apparently depend on what they have been eating.

The ruffed grouse of the United States is regarded by many as the finest of upland game birds and favored by the epicure. However, Mr. E. H. Forbush, in his monumental *Birds of Massachusetts and Other New England States*, gives accounts to show that in winter the ruffed grouse is known to eat leaves of laurel, which have poisonous properties, and that there are stories of serious poisoning resulting from eating the flesh of the birds. Such poisoning, Forbush points out, seems to have taken place only long ago and

only by winter-taken birds. Perhaps now that it is illegal to shoot grouse in the winter when they may have been feeding on laurel, such poisoning does not occur. This seems an additional reason for obeying the game laws.

Pigeons in the tropics are abundant both as to individuals and as to species and many are favored as food. However, Messrs. D. L. Serventy and H. M. Mitchell, in their recent volume on the birds of Western Australia, report that bronze-wing pigeons of two species are given to feeding on the seeds of the box-poison plant, and when they have been feeding on these seeds their entrails and bones, but not the flesh, are poisonous to dogs and cats. The effects of eating this poison seems to be that the dogs and cats have fits, become mad, bite at anyone within reach, and finally die in convulsions.

During Colonel Meinertzhagen's study of the birds of Mauritius he found that one of the pigeons there had a bad reputation from a culinary point of view. Reports have it that some of the people who have eaten the flesh of this pigeon suffered from extreme lassitude, while others reported the effects as convulsions. Strangely some of the people who reported sickness from eating this pigeon say it tastes well, while others who have eaten it without ill effects say that the flesh is bitter.

KINGFISHERS ON THE TELEPHONE



"WHAT COLOR is the kingfisher? Not the American one, but the European and Asiatic one? My husband is painting one and needs to know the colors," a lady's voice came over the telephone. I thought quickly. "Will it help if I explain the various kinds and colors of kingfishers and where they live? But no, lessons on taxonomy and zoogeography fall too flat most of the time." The lady's voice had a Central European quality. To her "the kingfisher" probably meant the little sparrow-sized kingfisher of the Old World scientists know as *Alcedo atthis*. So I'd better start with that. I described the cobalt-blue back, with darker wings, and dark bars on the crown; an earth-brown stripe through the side of the head, paling to whitish posteriorly, and with ochereous underparts.

"What color is the eye?"

"Brown."

"And the feet?"

"Red."

"And the nails?"

"Black."

She thanked me prettily. I tried to tell her about some of the other kingfishers, but she said no, she had enough, and hung up.

I sighed and thought regretfully of all the other things I had ready to tell her.

In the United States we think of the kingfisher as the belted kingfisher, larger than a jay, with a tousled crest and a voice like a watchman's rattle. But there are other species farther south in the Americas, and in the Old World there are still more. The tropics are their home. Only one species reaches Northern United States, and only one reaches Britain. But in New Guinea, for instance, there are about twenty-four of the ninety or so known kinds of kingfishers; the smallest tiny as a warbler, the largest nearly crow size.

Kingfishers, we call them, but many live on the dry land, and instead of catching fish catch insects or other tiny animals from the ground. One large species, with a broad shovel-like bill, is even reputed to dig in the earth to get its food of earthworms.

They all look much alike in shape. Once you overcome your surprise at seeing a kingfisher as big as a crow, or smaller than a sparrow, you recognize one anywhere—big-headed, large-billed birds with tiny feet that sit up quietly much of the time. Blue is a common color, but not all are blue. Some are generally reddish in color, some patterned with browns, grays, and whites tinged with blue. Many are decorated with crests, and a few species have elongated spatulate-tipped central tail feathers that have earned the species the name paradise kingfishers.

Its voice has given one species its name: the laughing jackass, the jackass kingfisher, or the kookaburra of Australia. "Ha ha huh huh ho ha ha huk" in a deafening chorus has been given as a description of its call. A. H. S. Lucas and W. H. D. Le Souëf, no doubt with tongue in cheek, record that "*on dit* that the jackass has been heard to laugh while a cicada [it had eaten whole] has been skirring inside him."

CLASSICAL ALLUSIONS Halcyon, Alcyone, and Ceyx appear in the scientific names of kingfishers. Scientific names make the layman shudder. Latin, he says, and if he's told they're not Latin, but rather Greek, it doesn't help any. But once you know the story of Halcyon (or Alcyone) and Ceyx, the names stick in your mind. In ancient times Halcyon was the daughter of Aeolus. And in grief for her drowned husband, Ceyx, she threw herself into the sea. The gods, out of compassion, changed both into kingfishers. Halcyon was

also used by the Greeks as a name for the kingfisher and it was fabled to make its nest on the sea, and to quiet the waves for its incubation period. Poets still use Halcyon for the kingfishers in reference to calm, happy, peaceful days, Halcyon days; the sort of days in which the kingfishers can nest on the quiet waves.

The lady had not waited for all this. She had gone. I would have liked to see the picture her husband was painting when it was finished.

ON IDENTIFYING SEA SERPENTS



THE LOCK NESS MONSTER reappears periodically in the newspapers. This monster seems to belong in the general category of "sea serpent." As a museum zoologist I've had little to do with such things. The stock in trade of a museum is specimens and if someone sends us a "sea serpent" (and I don't mean a water snake or a sea snake), we'll identify it. If it doesn't have a name we'll give it one and make a place for it in our classification. Until then we are aloof. We've had some little experience at times with "sea serpents" and the following will illustrate the sort of investigation and the results that we've had.

Years ago Sir Frederick Jackson was an administrator in East Africa. In addition to his official duties he was an enthusiastic and an able naturalist. So when a "sea serpent" was reported there he investigated.

IN KENYA The sea serpent was said to frequent Lake Naivasha in the Rift Valley of Kenya Colony. Up until 1909 there were many rumors of it, and Europeans had seen it with their own eyes. It always appeared on the lake about the same time each day, about five o'clock in the afternoon, always about the same distance from the shore, and was always traveling in the same direction, from north to south. All descriptions agreed that it was long,

black, and reptilelike, and that it kept appearing and disappearing on the surface of the water at short intervals.

Sir Frederick kept watch with one of the people who had reported it. And, sure enough, what appeared like a long black reptile appearing and disappearing, or like a school of porpoises, rising and disappearing, came into view. But Sir Frederick had binoculars and was able to make out that what to other people had been a long black reptile was in reality a long line of white-breasted cormorants in flight, on their way to their roosting quarters. As they flapped steadily along they were plainly evident, to the naked eye, as a moving black line; as they paused in their flapping and sailed on motionless wings they became invisible to the naked eye, though, of course, still visible through the binoculars.

IN NEW GUINEA Once, for a few startled moments, I thought I had a sea serpent before my very eyes. It was on the middle Fly River in south New Guinea. We were camped on a bamboo-covered bluff overlooking the river. Though about one hundred miles from the mouth, the tide made itself strongly felt here, and there was an abundance of driftwood. This driftwood, varying from freshly uprooted trees that had fallen into the river to waterlogged timber that had been long in the river, went up and down on the tide until it got out in the main channel and so on to the sea. One day at lunch, sitting in front of my tent, I was idly watching the driftwood. One piece in particular caught my fancy. Apparently it was the root of a partly submerged log, projecting about three feet above the water, and curved at the end so that it looked like the neck and head of a reptile with a casque on its head. Knowing it was a waterworn root, in fancy I even saw its eye. I called my companion's attention to it, as here was as close as we were ever likely to get to a sea serpent. Then, the "head" turned. It was alive. For a few startled moments it was a sea serpent. You can imagine our amazement at having a piece of driftwood that we had in fancy turned into a sea serpent come to life. Investigation became the order of the day. The binoculars that were constantly at hand were trained on it. The reality came as a further surprise. Our sea serpent was the head and shoulders of a cassowary which was swimming the river. Later I found that these large, ostrichlike birds, which have a large casque on their heads, are well known to swim, but I didn't then.

This seemed an ideal opportunity to collect a specimen. These birds may weigh up to 150 pounds. When shot in the forest there is the question of lugging them perhaps miles to camp. Here was one swimming up to our door.

We sat quietly waiting for it. But our native boys had seen it too, for next I saw them rowing the dinghy to it. An oar was brought into play to stun it. And then both the boys and ourselves found out something else. Dead cassowaries sink. When the bird was stunned by a blow of an oar, it disappeared below the surface and was never seen again.

CONSERVATION OVER THE TELEPHONE



RICHARD ORR, the *Tribune* reporter, called me one day about bronze grackles. It seems that the Chicago *Tribune*, in their "Day by Day on the Farm," had told about the grackles on the *Tribune* farm. A *Tribune* reader wrote in, expressing surprise that grackles were permitted on the *Tribune* farm and gave details of destruction by grackles of other birds, personally observed. What were the facts of the case? Should grackles be tolerated? Or should they be eliminated? Orr wanted to know.

This is the sort of question that is difficult. It is important, too, for it involves basic conservation issues. And there is no sharp, clear-cut yes-or-no answer. The question as to the grackle's character reminds me of the character of Moses, as explained when I was in school by a professor of the Bible: The black was there and the white was there; Moses was a character sketch in gray. And so with most creatures. They're both good and bad from our standpoint. Grackles certainly do kill other birds at times, and interrupt the nesting of some of our favorite songbirds. And yet, liking birds as I do, I tolerate them in my garden. On a trumpet vine on our garage in Chesterton, Indiana, one year we had a grackle build its nest on top of a domed English-sparrow nest. The young of both sparrows and grackles hatched about the same time, and the two families, within six inches of each other, were successfully raised without friction between the parents.

Quite evidently grackles are not always killers of other birds. As to robins or grackles being the "better" birds, if we had a robin's nest that we prized,

and the grackle killed the young in it, the grackle would be "bad." But if we were an inquiring farmer, and had to weigh the grackle against the robin, we might find the grackle "good" and the robin "bad." The grackle feeds its young vast quantities of insects harmful to the gardener; the robin sometimes seems to specialize in earthworms. Earthworms are beneficial to man, passing through the earth, making air and water more accessible, and, by passing earth and vegetable matter through their intestines, enrich the soil.

The house wren that warned the *Tribune* reader when the grackles were about is often prized as a garden bird; it is bold, saucy in appearance, and a vigorous songster. But it is also well known as a quarrelsome bird, prone to punch holes in the eggs of its neighbors, and it also may fill up with sticks nesting boxes so that other birds cannot use them.

The above was the gist of what I told Orr, and appeared in the May 5, 1950, *Tribune*.

Thinking of it afterwards, as is usual, I thought of many other things I could have said, and perhaps made more clear that no bird is all good or all bad, from our human point of view. Their relationships with the rest of the landscape are complex. I like to see butterflies flit about my garden. But butterflies are caterpillars at one stage. And caterpillars may eat some of the things in my garden. But some birds feed on caterpillars. If I eliminate the caterpillars because they eat the plants I like, at one stroke I eliminate the source of the butterflies I like, and food for some birds I also like.

Perhaps the partial answer, if answer there be in this imperfect world, is summed up by moderation: I can have some butterflies, some caterpillars, some plants, and some birds in my garden. If one becomes too abundant and interferes with the others, I prune it. Maintaining some sort of a balance, we can have some of each.

BIRDS WASHING FOOD

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WE NOT ONLY WASH ourselves and our clothes, but certain items of our food are regularly washed, as spinach, to get the sand out of it. Washing has been so important in our society that we've coined the term "Cleanliness is next to godliness." Possibly we've the snobbish idea it's a strictly human trait. Among other animals we don't expect to find water used for such cleanliness, and the raccoon, who does wash his food, is considered a sort of biological oddity.

But when we come to birds we find a surprising number of them that wash their food.

The dipper of our Western mountains in Oregon has been seen to wash insects and grubs before feeding them to the young birds. The parents held the food crosswise in the bill and the head was twisted rapidly from side to side in the water. Not until then was the food taken to the nest for the young.

The scene shifts to Africa. Four buff-backed herons were feeding on a flooded lawn at Gezira, Egypt. One of the birds captured a large insect, apparently a large black beetle. Holding the insect in the tip of its bill, the bird walked to the water, immersed the beetle three times, shaking and fumbling with it the while, and then swallowing it.

Then in Britain came a whole host of records, after an observation in Holland in 1946 of curlew sandpipers washing food. The birds were

probing the dry mud at the edge of a little creek. When one of the birds got a small sand worm, it at once ran with quick steps to the creek and stepped into the shallow water, where it dipped the worm a few times into the water before swallowing it. Then it trotted away for more. The editor of *British Birds*, the journal in which this was published, suggested that this might be a more common habit than the scanty published records would indicate, and invited observations.

A spate of records resulted in the succeeding numbers of the journal: a whimbrel washing crabs; a snipe, earthworms; godwits washing their food; with curlews it was reported to be normal; dunlins, greenshanks, redshanks, ringed plover, and oyster catchers were all reported doing this until it appears that with the group of birds we call shore birds—sandpipers, snipes, plovers, and their relatives—it may indeed be normal. The details of the observations strongly suggest that the reason for the washing, in many cases at least, is the same one that underlies our washing spinach; to get the sand and mud out of it.

HOW ANIMAL VOICES SOUND TO FOREIGN EARS



WHEN IN EL SALVADOR in 1951, I found that the common barnyard animals had much the same voices as the ones with which I was familiar in the United States. But when I saw their utterances written down it was another matter. The voices written in Spanish sometimes looked as different as the names of the animals written in Spanish. Take the donkey, for example (or *burro*, as they call it in Spanish). In English we call its "song" "Heehaw!" In Spanish they wrote it for me, "Aja! Aja! Ija! Ija!" There were a number of German scientists at the Instituto Tropical de Investigaciones Científicas, where I was working, and for comparison I asked them to write for me what the same animals said in German. The burro (*Esel*, they call it in German) says, "Ihå! Ihå!" in German. Despite the difference in the appearance of these words, when they were pronounced by the various nationalities they sounded very similar. Compared with the original assinine pronunciation, the Spanish version was awarded the prize for being the best rendition of the beast's voice.

The cat's "Miau, miau, miau" in Spanish, "Miau, miau" in German, and "Meow" in English were all very similar in appearance as well as sound. The duck's voice came out differently. In Spanish it was "Cuá, cuá, cuá," in German "Wack, wack," and in English the initial "Cu" or "Q" sound of the Spanish, and the final "k" sound of the German are united into "quack." The

hoot owl came out much the same in pronunciation, though it looked different in the Spanish "*Ju*," in German "*Hu*," and in English "Who."

The cow's, the pig's, and the frog's voices were also rather similar in the three languages: the cow's in Spanish being "*Meu, meu, muuu*," in German "*Mŭh, mŭh*," and in English "Moo"; the pig's "*Grup-grup, wink*," "*Óŭik, Óŭik*," and "Grunt, oink"; and the frog's "*Cruac, croac, croac*" "*Quak, quak*," and "Croak." The barnyard rooster has a difficult voice to transcribe in letters. In Spanish it was "*Quiquiriguiiii*," in German "*Kickeriki*," and in English "Cock-a-doodle-do." After listening to the various renditions by the various nations I could see how each rendition came into being, but as for deciding which was closest to the original I hesitated to choose.

When it came to the dog, the discrepancy was surprising: in Spanish it was "*Guán, guán, guán*," in German "*Wau, wau*," and in English "Bowwow." The German and the English are close enough. But though I went outside and listened to the dogs in Salvador, never did they seem to say, "*Guán, guán, guán*," though I must admit that neither did they seem to say, "Bowwow."

SIGHT IDENTIFICATION



SOMETIMES when I'm trying to decide whether the birds of the Cameroon Mountains of West Africa are the result of one invasion and variation *in situ*, or of two invasions, or whether the Himalayan red-billed choughs of Ladak are different from those of Nepal, or how the molt of the cassowary resembles that of penguins, I am called to the telephone to identify a bird someone has seen.

The chances are it's a starling. I've not kept a record, but I fancy half the questions are on identification of starlings. In the distance starlings are black, and people know them. But close up, where details can be seen, they puzzle people with their variety. The young may be dull brownish; the adults may be speckled in the winter; in the spring the speckled tips of the feathers wear off and they're all black. But the black is iridescent, and in sunshine glitters purple or greenish. And the bill color changes too: it becomes yellow in the spring.

Sometimes it's surprising how you can spot a bird from a brief description. Take this one: a bird that sits with its stomach on the ground, and has a big mouth, and long whiskers; a whippoorwill obviously. Or take this one: a bill like a chicken and with flat feet at the back; obviously a pied-billed grebe.

There was one that absolutely stumped me for a day. The lady said it had a bill like an eagle, and a tail that stuck up. For the rest she was vague. Often

habits, actions, or habitat are a help to me in placing a bird, but I could get nothing to help—not even where she had seen it. I admitted I couldn't help her. The next day someone brought in a picture puzzle out of a newspaper, and there, right in the center, was my bird. It was a dodo! We don't mind helping people learn things, indeed we consider it part of our job, but to help them work puzzles is too much!

MY LESSON Sight identifications of most students probably contain errors. On common species it's not important, as quantitatively they cancel out. But when a bird tripper, anxious to make a new record, wants me to help him decide he saw an exotic tern, I'm very careful—I've had experience. Rarities have to be checked on all points, not identified by elimination or on a few key characters. One of the best lessons of caution I had in New Guinea. It was in the mountains. Each morning I hunted in a forest where I'd found a new genus of bowerbird. Anything might occur, I thought. Then I saw flying through the treetops what could only be a magpie. A long-tailed, black and white bird, its pattern was unmistakable. There was nothing like it known from New Guinea. It would be an extension of range from Asia. Or it was a new and unknown species. Anyway I needed it as a specimen. But it was shy and eluded me. Morning after morning I haunted the forest. Finally I got the bird. And it turned out to be a partly albinistic specimen of a common, black, long-tailed bird of paradise. The abnormal white areas in its plumage had fooled me completely. But it helped teach me caution as to sight identifications.

One of my Gary friends, Mr. Raymond Grow, who is a keen bird student, has the proper approach, as his identification of a winter duck showed. There were a number of unusual winter birds that season (1951-52): brown-headed chickadees, pine and evening grosbeaks, and red-breasted nuthatches, all from the North, were present. It was the sort of winter one expects other rarities from the North.

DUCK CAUSES CONFUSION Mr. Grow had seen at the edge of Lake Michigan a duck he didn't know; it was boldly patterned in black and white, a big duck. An immature male eider seemed the only possibility. He came into the museum and we went over specimens, noting the difference in the shape of

the head between the king and the common eider. He studied the descriptions and the plates. Nothing quite fitted. Unsatisfied, he went back to Michigan City, found the duck again, and suddenly realized it was a muscovy duck, partly albinistic, and escaped from someone's barnyard.

It's not the first time a muscovy has caused confusion. Only a year or so ago we had a duck sent us from the Philippines that our correspondent wrote was shot swimming in a river with a Philippine mallard and surely represented a new species. But it turned out to be a muscovy whose original home is tropical American but has become domesticated and transported by man to far parts of the globe. Occasionally birds escape and take to the wild, even as this Philippine bird had done.

GREEN HUNTING JAYS TURN BLUE



SOMETIMES in "working out" a bird collection things get dull. In identifying the specimens, and writing down why they are this species, or that species, or subspecies, it seems routine; as though it were simply routine putting things in the categories ready for them.

Such was my feeling one day as I worked over Himalayan jays and magpies from Nepal. I'd done the yellow-billed blue magpie, and the red-billed blue magpie, which both fell into their places smoothly. Then I got out the literature, the pertinent keys, and descriptions for the next species, the green hunting jay. It's a beautiful, pale, apple-green bird, with a green crest, and set off by dark red wings. It checked with the descriptions, and I wrote *Kitta chinensis*, its scientific name, on the label. Then, to check the species' identification and to determine the subspecies, I turned to the collection, to the birds from India, Siam, and north Indochina, which should all be the same.

I pulled out the drawer—and blinked at the jays, rows of them; all pale blue with brown wings. I looked at the name on the case, on the tray, and the name on each specimen. They all said the same, *Kitta chinensis chinensis*, and it was the bird described as green, like my new specimen. It was uncanny. The new green specimens and the old blue ones were identical in size, in structure of bill, crest, feet, tail; they must be the same. And they were. The book, I found, described how the colors changed with age, and in

John Gould's magnificently illustrated folio, *Birds of Asia*, published in 1861, he had the green hunting jay depicted both as a green bird with red wings and, in the background, a "blue" green hunting jay like our museum specimens. When alive, and when freshly killed, the birds are green. But with the passing of time the green changes to pale blue, and the red wings to brown wings. Probably my new specimen, now a year old, is less green than it was when fresh. And when twenty years old, like our museum skins, it will be blue too.

The riddle was solved, and it fits into a well-known phenomenon, "museum age" or post-mortem change. "Foxing," we call it for short. We see it in the male American merganser, where the lovely rich salmon color of the fresh bird becomes plain white. The emerald cuckoo of Africa has vivid rich yellow under parts when fresh, and this too becomes dingy white. Gray Canada jays become more brownish. Birds that are olive or other shades of green tend to become more olive; brown birds tend to become more russet or foxy (hence the term "foxing"). We keep all our specimens in dustproof, lightproof metal cases. The change is not caused by fading. Apparently it's a change in the pigment, perhaps from oxidation.

Taxonomists, the men who classify and name birds, have been fooled by it. Old skins used to represent the birds of an area may give a quite different idea of what they are like than do fresh skins, and when skins of different age are compared, the conclusions may be wrong.

Foxing is one of the pitfalls for the unwary taxonomist, and something he has to guard against.

HOW BIRDS USE COWS AS HUNTING DOGS

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THE SPORTSMAN out for quail or woodcock uses dogs to drive out the birds for him. Starlings and cowbirds about Chicago use the same principle in hunting grasshoppers. Instead of dogs they use cows, though of course the cows are intent on something else and presumably unconscious of the fact that they're helping the birds.

As the cow grazes slowly across a meadow, it scares up grasshoppers close in front of it. The cowbirds and starlings take advantage of this. Instead of covering the meadow on foot, constantly alert for a sitting grasshopper, or to chase one they flush, the birds keep with a grazing cow. They take up a position by the head, or a foot, and catch the insects the cow disturbs. The cow is so much larger than the bird that it is likely to flush more insects. The grasshoppers on the wing are much easier to see than when at rest in the concealing grass, and some fly directly toward the bird. Too, the grasshoppers fleeing a cow are less likely to be alert to other dangers.

CONFIRMED BY OBSERVATION The advantages of this to the bird are obvious. But we've just assumed they were, and we had no data on the relative efficiency of the two methods of hunting. A few years ago, however, while

in El Salvador, I was able to get quantitative data proving that using a cow as a beater was advantageous, as we suspected, and showing how much more effective it was, something we did not know.

The bird concerned was not the starling, which does not occur there, or a cowbird, which occurs but consorts little with cows, but was the grove-billed ani, a black cuckoo about twelve inches long of the tropics of Central and South America. Like our starling and our cowbird, it kept with cows, catching the grasshoppers and other insects that flew up. Both anis and cows were common in the grassy fields about our headquarters in San Salvador. We decided, my son Stanley and I, to watch anis with cows for a few hours, and then without cows for a few hours; thus getting the average rate for each type of feeding. We quickly found it wasn't as easy as that. Something always happened; even on the levellest and most open fields the birds were constantly disappearing behind a tuft of grass, or in a hollow, or, if nothing else, behind the cow's head or feet. Then, too, the ani we elected to watch wouldn't pay attention to the job in hand. It would wander off, or go to sleep. And sometimes, when we were about to discontinue watching a somnolent bird, it would snap up an insect. Perhaps it had been watching all the time. Finally we found we had to record observations of many short periods, of from three to fourteen minutes each, and add them together.

By dint of much patient watching we got our data. In the dry season when insects were scarce and the grass short, it took an ani, hunting alone, two minutes on the average to find an insect. In the same length of time hunting with a cow the catch averaged three insects. Thus hunting with a cow as a beater was three times as effective as hunting alone.

The effect of the change of the season in abundance of food for the ani was very striking. In the wet season the grass began to grow fast, and insects became common. Then the anis had an easy time. Without a cow an ani averaged between three and four insects a minute, more than six times as much as in the dry times. There was less incentive to use a cow as a beater, with food so abundant, but when the ani did so, its rate of finding insects was still higher: between four and five insects per minute. In a table it looks like this:

Average Number of Insects Per Minute Found by Ani Feeding

WITHOUT COW WITH COW

Dry Season	.5	1.5
Wet Season	3.4	4.7

But the three-times-greater-results in a given time in the dry season do not tell the whole story as to the effectiveness of using a beater. When an ani was hunting by itself it walked about, covering a surprisingly large amount of ground. When using a cow as a beater, not only did it catch more insects in a given length of time, but it also walked about much less, saving a great deal of energy.

This is not true co-operation between cow and bird, for they're not working together toward a common end. It's not exploitation of the cow by the birds, for the cows lose nothing. It is closer to a form of harmless parasitism, for the ani profits from the activities of the cow without either harming or helping the cow. It also illustrates how sharp birds are—ready to take advantage of any factor in their environment that will help them get their food.

EARLY BIRD LISTING



I WONDER how many of the people who go out making lists of spring birds know that bird listing goes back to ancient times. It's a modern sport, but earlier bird watching was serious, and a competitive listing of birds played a part in as important an event as the selection of the site of the city of Rome.

The story, as Plutarch tells it, is that Romulus wanted the city on what became known as Roma Quadrata; Remus wanted it on the Aventine Mount. As was the custom in those days, they concluded at last to decide by a divination from a flight of birds. The twins placed themselves apart at some distance and watched. Remus, they say, saw six vultures, a truly notable flight; Romulus saw twelve and from this rare and unusual occurrence Romulus' choice of the site for the city was accepted.

VULTURES HIGHLY REGARDED Partly from this the vulture became chiefly regarded by the Romans in their divinations from birds. But even before this the vulture was highly regarded. Hercules, it was said, was always very joyful when a vulture appeared to him upon any occasion. He considered it the least harmful of creatures; not pernicious to corn, fruit tree, or cattle, it never killed or hurt any living thing. It was also thought not to eat other birds, a weighty point in its favor, as Plutarch quotes from Aeschylus, "What bird is clean that preys on fellow bird?" And apparently its deciding

claim to esteem was its rarity and infrequency, which gave rise to the opinion in some that it came from another world, an opinion foisted by the soothsayers of the day.

Earlier yet, birds played a part in Rome's history. Plutarch warns that some give you mere fables of the origin of Rome, but it is widely current that Remus and Romulus, fathered by Mars, the God of War, were exposed in a remote place to perish. This would have taken place, but for a she-wolf that nursed them, and birds of various sorts that brought little morsels of food which they put into their mouths. Some, however, hold the belief that not birds of various sorts but a woodpecker was the bird that constantly fed and watched the twins, and even in Plutarch's time the Romans still worshiped and honored the woodpecker for this service to the founder of the city.

BATTLE OF THE SEXES AND ITS EVOLUTIONARY SIGNIFICANCE

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I USED TO THINK that the battle of the sexes so ably portrayed by James Thurber was artificial, a man- and/or woman-made thing. But recently I've come to see it as old—probably as old as sex itself in the animal world.

Under the severe tide, "Secondary Sexual Characters and Ecological Competition," in a paper from the Bird Division of the Chicago Museum, I've outlined the possibility of competition for food, between the sexes, being a factor in evolution, responsible in part for characteristics of structure and traits that distinguish them.

In circles that discuss evolution the idea is current that food competition is important between species. It may even be stated as a rule: two species with the same food habits cannot live in the same place. Competition drives one out, unless they have different food habits. These differences seem especially evident when you look at closely related species, and they are accomplished in a variety of ways. A habitat difference is very common. The long-eared owl hunts in the woods—its cousin, the short-eared owl, hunts the meadows; the song sparrow favors the drier shrubbery while its cousin, the swamp sparrow, lives in wetter shrubbery.

THE SIZE FACTOR Sometimes the difference is accomplished by size; take the downy and hairy woodpeckers of our wood lots, very similar except that one is larger and is adapted for larger prey, the other smaller and adapted for smaller food items. Sometimes they feed differently, as the Baltimore oriole, which picks flowers and pecks through their sides, while the orchard oriole probes into flowers as they hang on the branches. Thus more individuals of several species live in an area.

When a pair of birds "sets up housekeeping" and starts "raising a family" they can no longer drift about, looking for easy living and places where food is plentiful. Their wanderings are restricted by having a fixed point, the nest, as their center of interest. Two individuals must draw on the food supply from an area about the nest. Competition would be extreme, and, if there were a scarcity, perhaps critical.

We know how different the sexes may be; how different the rooster is from the hen in our domestic fowl, or the drake and the duck in the mallard, or the red male and the green female of the scarlet tanager. These sexual differences have mostly correlated with display and mating. But logically there should be differences in feeding behavior and adaptations between the sexes.

The basic idea is contained in the old nursery rhyme:

Jack Sprat could eat no fat,
His wife could eat no lean;
And so between them both,
They licked the platter clean.

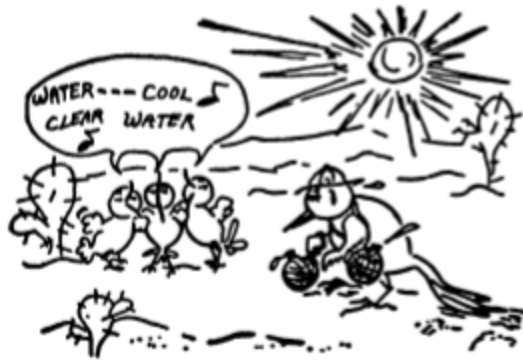
The two birds of a mated pair, limited to a single area, could be expected to have different food preferences or adaptations for getting it. And we find that there are cases of this. The most striking is that of the huia from New Zealand, of which I've written in a Chicago Museum bulletin. Both sexes have similar food preferences, especially wood-inhabiting insects, but they get them in different ways. The male has a short, straight, stout bill for digging out the wood-boring grubs, woodpecker fashion; the female has a much longer, slender, and curved bill for probing into holes for them, creeper fashion. The female may get grubs in wood too hard for the male to chisel. They supplement each other.

DIET VARIATION BY SEX It is possible that further study may show more sexual differences to have a feeding advantage; the larger size of female hawks fitting them to take larger prey; the smaller size of certain female songbirds fitting them for smaller prey, the smaller bills of female hornbills, the straight bill of the male western grebe, and the upturned bill of the female. Perhaps all are of advantage to the species in giving each sex slightly different advantages in getting food.

Selection could have its effect in the populations with most sexual difference in feeding habits being most successful in raising and leaving progeny. Thus, slowly, differences between the sexes would accumulate. However, it must be kept in mind that this sort of evolution would be limited. The drifting apart of the sexes would be checked by the necessity for their coming together periodically for at least a short period, at nesting time.

WATER IN THE DESERT

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WATER is a precious thing in the desert. Without it no life is possible. When rains come plants spring into vigorous growth. During the long stretches without rain they rest, some as seed, while some plants store water in root systems, or in large trunks. Animals have developed a number of ways of surviving long dry spells in arid country.

Among mammals the kangaroo rat of our Southwestern desert seems able to get along without water. This is caused by an arrangement within the body whereby the necessary water is manufactured within the animal from other foodstuffs: metabolic water.

The accessibility of drinking water in a desert may be the determining factor in whether or not some birds can survive there. The nests of Gambel's quail must be close enough to drinking water for the newly hatched young to walk there, else they perish of thirst. It has been said that newly hatched chicks of the related valley quail of California cannot travel more than a few hundred yards from their hatching places without water. Broods hatched farther away are doomed to die.

Sand grouse, relatives of the pigeons that have adopted the general appearance and habits of quail, live in the Old World, primarily in arid or even desert areas. Where they occur their daily traveling to water is a well-marked phenomenon. Their flight is swift and powerful, and though they

may traverse long distances of barren, inhospitable country to watering places, their punctuality in arriving at water, morning and evening in some species, is remarkable.

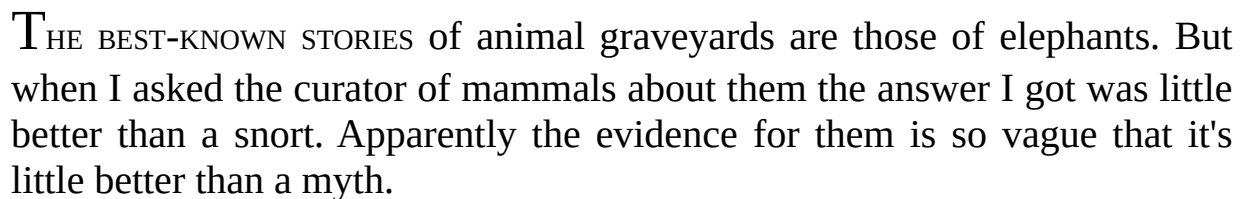
But what of the young of these desert dwellers that need water? A most unusual situation exists; indeed it seems to be unique. The old birds bring water to the young! This has long been recorded, but as recently as 1921 it has been questioned. However, Mr. Meade-Waldo's observations on birds in captivity seem to definitely establish the custom, and its methods.

PARENTS CARRY WATER Both birds incubate the eggs, the male by night, the female by day, and both parents care for the young. But it is the male only that brings water to the young. He rubs his breast violently up and down on the ground, and then, his feathers awry, he gets into his drinking water and saturates the feathers of his under parts. Then, in captivity, he would run to the hen, make a demonstration, whereupon the young would run out from under her, get under him, and suck the water from his feathers. This they did by passing the feathers through their bills, continuing and changing about until the supply was exhausted. It was found that until the young can fly they take water in no other way.

This was in captivity. Presumably in the wild the process is the same, the adult flying with wet under parts from the water hole to the resting place where the young are under the care of the female.

The similarity of the young sucking water from the feathers to young mammals suckling their mother has been pointed out. But another and a truer similarity exists: that of the young sand grouse getting water from the feathers, and young quail getting water from dew-wet leaves in areas where dew is heavy and there is but little surface water.

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In the antarctic Dr. Robert Cushman Murphy found on the island of South Georgia a place where Johnny penguins went to die. It was in a lake in a coastal range of hills. The lake bottom was thickly strewn with scores of penguin bodies, all of which had apparently died a natural death. The icy water, Murphy thought, might preserve them for years. The hills, away from the sea, seem a surprising place for the graveyard of such aquatic birds as the penguins, but it correlates with another peculiarity of their mental makeup. They like to nest on high land, or at least far from the sea. The blind instinctiveness of much penguin behavior is well shown by these birds when there is no high land on their nesting island. Then they may nest so far from the beach on which they land that they are close to the water on the other side. Yet they always returned to the sea by the long route, never taking the shorter route.

Another aspect of this preference for land distant from the sea is shown by their behavior when threatened with danger from man or dog. They flee away from the sea, back onto the land, when safety for them actually lies in the sea. Presumably this fixed behavior dates back to the time when the seal that is called the sea leopard was the penguin's main enemy. Then the sea held their only danger. With man's arrival the situation changed, but only after considerable experience with man do the birds change this behavior.

Apparently, when the time came for the penguins in this South Georgia graveyard to die, they followed their age-old pattern, climbing to the high country and away from the sea.

IN A HOLLOW TREE In a hole about eighteen inches in diameter and twelve feet deep in the trunk of a wych elm in Hants, England, Ursula M. Grigg reports finding the bones of at least ninety jackdaws, thirteen starlings, six green woodpeckers, and twenty-five stock doves. All the remains were clean, and not much broken or decomposed. The idea that these bones were the remains of owls' or other predators' feasts was discarded for a number of reasons; as was the idea that this had been a natural trap, the birds entering to roost or nest and being unable to escape. The most tenable idea seems to be that this was a favorite roosting place in winter, and that during the severe weather old and weakened birds, roosting there, succumbed and added their bodies to this communal grave.

ON AN ISLAND Another instance comes from the little Cape Verde Isle of Cima in the South Atlantic. A photograph in the *National Geographic* magazine for 1927, Vol. 52, P. 27, has the caption that this island is unique and uninhabited and covered with the tiny bones of millions of petrels which in ages past have come here to die. Certainly the plate shows an amazing litter of bird bones on the tiny plateau of this islet.

Petrels are mostly pelagic birds, coming to the land only to nest on isolated islets. Can this "graveyard" be merely the normal accumulation of the bones of the nesting season mortality, or can it be that the birds actually come here to die?

ANIMAL GARDENS

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BEST KNOWN of the "gardens" and "animal husbandry" of the lower animals are those of the ants; the aphids kept by the ants for the sake of a sweetish secretion, and the underground fungus garden of the ants. In the vertebrates I know nothing comparable to this, but we do get a number of cases where there is a definite relation between the animals and the growth of vegetation.

It has been said that in the antarctic the nesting colonies of some penguins are detrimental to the vegetation. The constant passing and standing of the birds on the limited areas of soil preclude the growing of vegetation over sufficiently large areas to be an important factor in hindering plant growth. But the reverse is true of the Johnny penguin in the Falklands, where it is sometimes known as the best farmer in the country. The Falkland Islands, off southern South America, are cold, wet, and windy. Sheep raising is one of the main industries. And the Johnny penguin helps to provide better pasture for the sheep. The birds nest in colonies and their droppings help to enrich the land so that the grass grows taller and richer. Rather than using the same area for their breeding colony each year the birds select a new, clean area at the beginning of each breeding season, so that they improve the ground over a larger area.

From the arctic comes another example of a relationship between bird and plant. On the arctic barrens, here and there, are large boulders, erratics left

by the glacier that covered the land in times past. And on these boulders, and here only, one finds patches of bright yellow or reddish lichen known to scientists as *Xantheria* or *Xanthoria*. Apparently its presence is owed to the fact that these boulders are the lookout places of snowy owls, hawks, and other birds. Their droppings, falling on the rocks, provide the nutrient layer necessary for the growth of the lichens. It is probable that these lichens are transported from place to place by the birds carrying the soredia on their feet. In recognition of the close relationship between these lichens and birds an ecologist has coined the rather formidable term "ornithocoprophilous" to express the relationship.

Also in the arctic are the arctic-fox gardens. The arctic fox often makes its burrows in sandy places, and about the entrance to the burrow accumulate remains of former meals, fox droppings, and suchlike animal debris. This in time enriches the soil and the vegetation there grows taller and more lush than elsewhere on the barrens. This lush vegetation attracts the small, mouselike arctic rodents, the lemmings, that feed on green, succulent vegetation. There is of course one further step in this chain. One of the important foods of the arctic fox is the lemming, which he thus brings to his door by the richer vegetation he unwittingly causes to occur there. A charming arrangement, one of the old naturalists called it.

DROPPING THINGS

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THE STORY is well known, being recorded by Pliny, of how the poet Aeschylus came to his death through a bird mistaking his bald head for a rock and dropping a turtle on it. The bird was evidently the lammergeier or "lamb vulture," one of the largest and most magnificent of the Old World birds of prey; nearly four feet long. In the Atlas Mountains of North Africa its normal food is turtles, and these it cracks open, so that it can get at the meat, by carrying them up into the air and dropping them on a rock. Now it lives in the Himalayas and in Africa, having been almost if not completely exterminated from Europe because of its alleged predation on sheep. Not only turtles but bones are treated in the same manner, to get at the marrow. Though the habit is well known, it is surprising how difficult it is to find a firsthand description of it. So far I know of only one description written by an eyewitness. And yet, in East Africa recently a stony mountaintop was found littered with broken bones that seemed to be the result of the lammergeier's habit.

GULLS DO IT As I have mentioned, gulls open clams and mussels in this way; and crows, which are among the most intelligent of birds, do it also. They pick up the mussels left exposed by the falling tide, fly up above a hard stretch of beach, a big rock, or a stretch of nearby paved road, and drop the

shellfish there. While in general this practice is restricted to a few groups of birds, it is practiced by them in many far parts of the world. The Pacific gull of Australia, widely separated from its near relatives, has the same maneuver for opening shellfish as has our herring gull.

It's hard to understand just how this habit came about. One can imagine that some birds found it out by accident when flying about with a stubborn "nut" they were unable to crack. Or perhaps it was in play they found it. The raven is known to fly about carrying and dropping things in play.

SPARROWS DO IT TOO Often, to find a background if not an explanation of a habit, we look about to see if it's used in some other connection. I've already mentioned the play of some of the crows. Only one other "dropping" habit has come to my attention, and that is a single record for the very common house sparrow. Edmund Jaeger writes that in Nebraska, and again in Riverside, California, he saw house sparrows on gravel roofs, dropping small stones over the edge. The pebbles, or small bits of crushed stone, were carried to the edge of the building by the sparrows, dropped, and as each pebble was dropped the sparrow turned its head, apparently the better to watch or listen to the pebble fall and strike. No obvious utility appeared in these actions. It, too, looked like pastime. Perhaps there was no better reason behind them than that behind small children dropping stones down a well.

LEARNING BY BIRDS

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OF COURSE birds can learn. Indeed there's a trite saying that no animal has been discovered so low that it cannot learn. One of the simplest cases of learning is shown by parts of some experiments I carried out years ago on the curve-billed thrasher. I had raised a number of these thrashers by hand, and in connection with finding out about their tasting abilities I first fed them on the white of egg, hard-boiled and cut into little squares. They liked it. Then I soaked more squares of boiled egg white in evil tasting (to me) formalin. The birds came to the dish, and also ate them. But after that for a week they refused to eat such egg white. They had quickly learned to avoid the ill-tasting food.

BAD-TASTING FOOD Once I hand-raised a barred owl from a nestling to adulthood. Sometimes getting food for it was a problem, and upon occasion I fed it frogs, which it seemed to like well enough. But then came a day when I fed it a toad. The owl seized the toad at once. Now toad skin, presumably as a defense weapon of the toad, secretes a substance irritating to the mucous membranes of some animals. And this was evidently irritating to the owl, for it did not hold the toad long in its bill. It spat it out, and the owl's face gave evidence of disgust. After that the owl not only refused to take toads, but it also refused to take frogs such as it had found

palatable before. Evidently frogs looked too much like toads. The learning was effective, and extended not only to the original object, but also to other, similar objects.

BUCKET-DRAWING When in Florida with the Archbold Expeditions I was studying blue jays. A very simple but amusing thing that chickadees learn is to sit on a perch and pull up a little container of food that dangles far below the perch on a string. Jays, along with crows, are among the most clever of birds, as I've said before, and I gave two jays in one cage a chance to learn the trick. In three days one of the jays was regularly and quickly pulling up the little bucketlike container and getting its food from it. The process was simple: the jay reached down, seized the string in its beak, secured the slack under its foot, and reached down again for another pull. Sometimes five separate pulls were needed to raise the food bucket the eight inches to the perch.

The jays were regularly fed in this manner. Soon I noticed that only one of the two birds pulled up the bucket, though the other also fed from it. In effect one was depending on the work of the other. After this had gone on for a month, I wondered if the second jay, which had never done any of the work, would be able to pull up the bucket if left alone. Certainly it had had lots of opportunity to learn by seeing its cage mate go through the motion. So I left it alone in the cage. This second jay, despite its chances to learn by observation, took one day longer to learn how to pull up the bucket of food than had the first jay. The jays certainly learned the trick quickly through a trial-and-error process, but simply watching the process seemed to be of little help in learning it.

CAN BIRDS COUNT?

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IF BIRDS can count, it's a rather rudimentary thing—perhaps no more than impressions of the size of groups. The widely known example showing that birds don't seem to distinguish between one and two persons is the ruse used by bird photographers and students of birds who are using blinds from which to watch the birds at close range.

The hide, or blind, is a little hut built perhaps a few feet from the nest to be photographed. If the photographer enters the blind in the sight of the parent birds, and conceals himself there, the birds who saw him go in will be a long time in coming to the nest and in resuming their normal activities. But if the photographer takes a companion with him, both go into the blind and conceal themselves, and then one of them goes away leaving the other concealed, the bird quickly disregards the intrusion and goes about its activities as though no one were left in the blind. This subterfuge has long been used and is very successful. Apparently the bird is unable to distinguish between the two people that arrive at the nest and the one only that leaves, and behaves as if both had gone away.

In my duck-hunting days a duck hunter who used wooden decoys told me he was sure that there was a certain number of decoys necessary before they were effective.

The decoys were wooden blocks, carved and painted to resemble life-sized ducks, weighted to float like them, and anchored in shallow water in a flock

within gunshot range of the blind in which the duck shooter sat. The idea was that ducks flying by would see the flock on the water, assume that here was a safe resting place, and fly in and light, or attempt to light among them, giving the wild-fowl gunner an opportunity to shoot the wild birds.

The duck shooter claimed that if less than twenty-five or thirty decoys were put out in the flock, the setup was much less effective than if more than twenty-five to thirty decoys were used. He thought that the ducks could distinguish between less than twenty-five or thirty and more than twenty-five to thirty, and favored the latter. Though this is distinguishing between greater and lesser amounts, it hardly comes in the category of counting.

DISTINGUISH "MORE" FROM "LESS" However, a series of experiments summarized on Page 121 in the periodical *Bird Banding* for 1940 seem to indicate that birds can distinguish between different numbers of things, such as peas and numbers of dots. The birds, including pigeons, parakeets, and jackdaws, were trained either to choose a certain number of objects under certain circumstances, or to choose between two quantities of objects with reward and punishment motivation. It was found that these birds were able to distinguish up to a maximum of six. That this is really counting in the human sense of the term, which is linked with speech or written symbols, is improbable, but it does indicate, as one would expect, that birds do at times distinguish between different quantities, and sometimes with considerable precision.

COURTSHIP FEEDING

[\[Ref\]](#)



A YOUNG MAN, giving his best girl a box of chocolates, and a bird, giving his prospective mate a worm or a berry, have this in common: they are both practicing courtship feeding. Further, humans and birds are the only vertebrate animals that do this.

With birds, during courtship, the female often begs to be fed by acting like a young bird—with fluttering wings and widely gaping mouth. The male normally places the food he has collected directly in the open mouth of the female.

The significance of this courtship feeding has been discussed especially by David Lack, in the scientific journal, the *Auk*. It seems that in courtship feeding the food as such is not of primary importance. The female does not need the food she is begging for; indeed she may have had a full meal since her mate, whom she is soliciting, had last eaten. Perhaps it is of help in maintaining the bond between the pair during the period that exists before they have a nest and young to look after. In this connection it is interesting that with waxwings during courtship feeding the fruit that the male gives the female may be "handed" back (by beak) and the food exchanged back and forth.

In looking for significance and correlations in courtship feeding we find that some species practice courtship feeding and some do not. And the birds

that do practice it are usually those in which both sexes care for the young. It might be considered an early, useless appearance of a habit that later becomes useful when the male feeds the incubating female and helps feed the young.

This type of behavior, in which an act used elsewhere is introduced into courtship, is sometimes called "symbolic." Other such symbolic acts are the preening that sometimes takes place between a pair of mating birds, and the passing or the manipulation of nesting material long before there is a nest to be built.

Some species during courtship go through actions that resemble courtship feeding except that no food is passed; the bill touching of the mourning dove and of the waxwing falls in this category. Perhaps it is incipient courtship feeding on its way in the long course of evolution, either upward, to include food, or downward, away from courtship feeding.

Their functions seem to be to give something for the pair to do; something they can share. It helps fill up the pair's day and keep them together. It is something that helps strengthen the bond between them, against the day when both will be working together raising a brood.

THEY TURNED THE TABLES

[\[Ref\]](#)



MOST BIRDS PREY on animals enough weaker than themselves to be in no danger from their prey; their hunting is more like that of the gunner after rabbits than that of the hunter after lions. But there are exceptions.

The great blue heron, armed with a spearheadlike bill, lives largely on fish. These it spears in the water, stalking about after them on its long legs, or waiting like a bird on a Japanese screen, as patient as any fisherman, for its prey to come within striking distance. The heron's size and its great bill render it safe from most enemies. But it sometimes overestimates its ability. Audubon recorded one on the Florida coast that, standing in deep water, up to its belly, struck a fish too large for it. The fish dragged the bird for several yards, now on the surface, now underwater. Finally, after a severe struggle, the heron freed itself. It was exhausted, and stood near the shore, head turned away from the sea. As if, Audubon said, it was afraid to make another attempt at fish catching.

A more serious encounter for the bird was recorded in *Field and Stream* magazine. The heron had caught a shad about a foot long. He tried to swallow it, but it was too big to go down. He tried to disgorge it, but the fins of the fish, acting as barbs, kept it from slipping backwards and out. The result was death for both bird and fish.

CAUGHT BY A CLAM The oyster catcher, a large black and white relative of the sandpiper, feeds on, among other things, shellfish. Mussels and oysters look like hard nuts to crack, even with a stout, wedgelike bill such as that of the oyster catchers. The oyster catcher's favorite feeding times are when the tide has fallen and the shellfish are first exposed to the air and before they have closed up their shells, and again when the tide is rising and the shells are just beginning to open. The oyster catcher stabs into the shell, and with its bill cuts the strong adductor muscles that hold the shells of the bivalves together. The rest is easy. But a danger lurks here: what about stabbing into too big a shellfish, or making an inept stab? And this very thing has happened. On the South Carolina coast Mr. W. P. Baldwin found a trapped, drowned bird. It was held, with the tip of its bill caught in the shell of a hard-shell clam, as if in a trap. Apparently the rising tide had flooded and drowned the bird.

The raven eats most anything, living or dead, and except for man has little to fear in the northern forests where it lives. Yet from Wisconsin comes a record of one that met his death through a porcupine. The porcupine's quills are a dreadfully prickly covering that one would think would protect it from most encounters. Yet one animal, the fisher, kills and eats it as a matter of course, and wolves and bears sometimes eat them without too many ill-effects from the spines. The slow-moving porcupine has little regard for automobiles, and many are run over on country roads. A porcupine is too big and tough for a raven to kill and the Wisconsin raven had probably fed on a dead porcupine. Stuck through its gizzard was a quill, and another, which had apparently caused its death, was stuck in its heart, having apparently worked there from the digestive tract.

Many small insectivorous birds eat spiders as well as insects. This they do almost with impunity in temperate latitudes, where only occasionally do spiders make webs strong enough to trap a bird. But in the tropics, where there are more large spiders, their webs must be a greater hazard to birds. That the hazard exists in both climes, however, is shown by a goldfinch reported caught in a spider's web in Cape Cod, Massachusetts, and a dusky flycatcher caught in a spider's web in Cameroons, West Africa.

SURVIVAL OF THE UNFIT

[\[Ref\]](#)



TO CARE FOR the weak, the unfit, and the cripple is usually considered an extremely highly developed altruism in our society. As our society progresses, more and more provisions are made for the unfit.

In nature the unfit usually is soon weeded out. If an animal is unable to feed itself it is doomed; or if it is less successful than its fellows it has less chance of leaving progeny. That is natural selection.

Hence on both counts it comes as a surprise to find two well-authenticated cases of crippled birds, unable to search for food for themselves, surviving for long periods.

The first is T. R. Peale's record in 1848 of a brown booby on Enderby Island in the Pacific. An adult bird whose plumage indicated it was several years old was found on the island, and it had only one wing, the other having been lost by some accident and the wound completely healed. The bird was unable to go to sea and get its own food, and was being fed by its fellows.

The second was a frigate bird, found on the Revillagigedo Islands, reported by A. W. Anthony in 1898. This bird, too, was fat, and had one wing withered and useless, evidently from hatching. It had never flown. Frigate birds are masters of the air that snatch their food on the wing from the surface of the water, and a flightless frigate bird would be as badly off as a flightless swallow. The cripple had been fed all its life by its neighbors.

At first the uncritical might think, What altruism, what charity, for the healthy to feed these two cripples. But an explanation involving less advanced principles, principles more in keeping with what we know about bird behavior, is possible. Remember that young birds that are unable to begin feeding themselves at the proper time may continue to beg for food, and be dependent for a long time, as I have shown with young shrikes under the chapter, "Conditioning in Birds." Remember that a young bird begging for food may be fed by adults, not its parents, and even by other young birds (shown in "Bird Helpers at Nesting Time"); and we have the clue.

The cripples, hungry, begged for food; the healthy birds responded by feeding, as they might do to other begging young, and owing to the unusual circumstances both were continued.

These certainly are cases where the unfit survived. Natural selection has not operated. But such cases are rare exceptions.

DUST AND SNOW BATHING



THE TAXIDERMIST preparing a bird specimen for the museum sometimes has to deal with one whose plumage is soiled or stained. He may have to wash it with water. Then, to dry the plumage, fluff it, and help in arranging the plumage so it will lie smooth and natural, he may use a powder: corn meal, sawdust, plaster, or plaster and potato starch may be worked into the feathers, then dusted out again. It is interesting that birds themselves use and have used long before taxidermists a similar method of using dust in dressing their feathers, a fact that anyone who has watched domestic hens for any length of time must be aware.

A DIRTY BATH Recently I watched a house sparrow dusting by the railway track in the city of Chicago. The dust may have been in part "clean" earth, but in part it was soot, city dust, and soft-coal debris. The sparrows here were dingy, all had their plumage heavily impregnated with city grime, and looked very different from the sparrows in the country. And this sparrow I was watching when it had finished dusting was the worst of the lot. These city sparrows, even when they bathe in water, seem never to get much of the grime out of their feathers.

This reminded me that Oscar Heinroth once wrote that birds do not bathe to get themselves clean, but bathe as an aid in bringing their feathers into

order and making them lie smoothly. Perhaps he is right. Certainly my sparrow did nothing to clean himself.

It is in arid countries, plains and deserts especially, where many of the birds take only dust baths. In more humid regions water bathing is the rule. But some birds do both, like our flicker and our house sparrow, bathing now in water, now in sand.

In northern climates, when the land is held in the grip of winter, the water frozen over, and the earth covered with snow, neither dust nor water bathing is possible. Then, it has been recorded, some birds find a substitute in snow. Among other cases, in Alaska the hawk owl has been seen to perch in the snow on the tops of telephone poles, and go through the motions of bathing; in England a rook was recorded as bathing "in crisp powdery snow as if it were taking a bath in dust or water"; and in New England in midwinter juncos have been recorded bathing "in light dry snow, just as other sparrows take dust baths in hot weather."

The snow evidently is used as a substitute for dust in these northern latitudes.

DECORATION IN THE HOME

[\[Ref\]](#)



TO USE a bunch of flowers or a spray of leaves in decorating a room in a house is a refinement of civilization. As the flowers fade, or the leaves wilt, they are replaced with fresh ones. Sometimes a winter bouquet is used that will serve for months.

There are several birds that habitually deck their nests with green vegetation, and when it is wilted, it is renewed with fresh. The reason is not clear. It has been suggested it is to supply humidity and, by evaporation, coolness; it has also been suggested that its use serves a sanitary purpose. But whatever the reason in birds' eyes, it looks like decoration to human eyes.

This habit is common with a number of different hawks: for example, the red-tailed hawk is reported sometimes to have its nest, a bulky flat, basin-shaped structure in the crotch of a tree, "profusely and beautifully lined with fresh green sprigs of white pine, which are frequently renewed during incubation and during the earlier stages in the growth of the young." The golden eagle is said to add green grass, or green leaves often attached to the twigs from time to time to the lining of the nest, especially after the young are hatched; and the broad-winged hawk is said to add green leaves to the lining of its nest. In quite another group of birds the same thing also occurs. A carrion crow's nest in England was visited periodically from March to

August. Strangely no eggs were laid during this whole period, but the birds remained in attendance. When found, fresh sprigs of oak leaves were interwoven around the rim of the nest. On subsequent visits the oak leaves were found to have been replaced with fresh ones, and the leaves were kept fresh until late August.

The purple martin supplies another example. The nest boxes we put up for them supply their main breeding places in some areas. "The parents have a habit of collecting many green leaves and placing them in the nest, a practice which may tend by evaporation to reduce the heat" in the nest box. "Where large colonies are breeding they sometimes injure pear trees by stripping certain branches of their leaves," according to E. H. Forbush.

A Madagascar weaverbird provides an example of decorating the nest entrance of a quite different type of nest; in this case the nest is in the shape of an inverted retort, with the entrance through the spout. The entrance is decorated with green grass heads or with green leaves, and the males keep adding fresh green decorations even when the eggs are being incubated by the female.

It seems hard to believe that this is really decoration, that it is not for some purpose—either connected with the raising of the young, or more probably a leisure or substitute activity—something to keep the bird busy and strengthen the bond between bird and nest when it is not otherwise directly occupied with nesting activities.

CURIOSITY IN BIRDS



BEING UNABLE to ask birds questions that will receive answers, we have to judge their motives from appearances. And from the way some birds act curiosity seems a strong motivation at times. They show a disposition to inquire into things, especially strange things.

Young blue jays that I've raised and studied are among the most prying, investigating, inquisitive birds I've known. When well fed they devoted much time to examining things. Humans, of course, would examine objects by picking them up in their hands, looking at and feeling them, perhaps tasting them. The jays, with more limited equipment, would examine them with bill and eye. When the jays were very young their toes interested them. They pecked at and twisted their own and their neighbor's toes. Pencils and crayons on my desk appeared to interest them particularly. These they were continually pulling about and pecking at. They went about picking at lines on paper, knotholes in the walls of their cage, the red letters printed on a bottle label, and the buttons on our clothes. Cigarettes they liked to investigate by pulling them to pieces. It looked as if the jays were interested in finding out about the things around them by touch and taste as much as they could.

LURED INTO DANGER Compared with jays, ducks seem rather stolid creatures, but they have curiosity too. This was well known to the old-time duck

hunters who capitalized on it in duck shooting. The technique is known as "tolling" and I've used two variations of it in museum collecting.

Once on a little mountain lake in New Guinea I found a pair of ducks of a rare species I especially needed for our collection. I stalked them to the farthest bit of cover I could reach, a tussock of grass on the lake margin, behind which I lay concealed. But the ducks were still too far away for me to reach, and their feeding did not seem to be drawing them nearer. I remembered the gunners' trick of tolling, and tried it. I took out my white handkerchief, held it above the tussock of grass while I kept well hidden, and waved the handkerchief back and forth. The response was surprisingly prompt and gratifying. The two ducks turned at once and swam right in to me so that I secured them without any trouble.

Once on a lake in central New York State there was a flock of scaup ducks swimming well offshore. It looked as if they never would come in near the bank. Quite by accident a setter dog that accompanied us began to cavort along the beach. Again the ducks turned and the whole flock came swimming in. Only then did I remember that among old-time gunners there was the practice of using a dog thus, a dog that was even trained for the purpose, to jump high and run about very conspicuously while the ducks were far out, and as the ducks came swimming in, to keep lower and frisk about partly concealed so that the ducks would have to come close to satisfy their curiosity.

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